

TATA INSTITUTE OF SOCIAL SCIENCES
MUMBAI
SCHOOL OF DEVELOPMENT STUDIES

DS 27 – GIS and Spatial Econometrics in Development Research

SPATIAL PATTERNS OF DEMOGRAPHY AND WORK IN MAHARASHTRA

A district-level GIS and spatial econometric analysis of Census 2011

Mapping, clustering and modelling urbanization, household size, sex ratio, literacy and work participation across 35 districts

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Spatial Patterns of Demography and Work in Maharashtra: A District-Level GIS and Spatial Econometric Analysis of Census 2011

Abstract

State averages hide large differences between Maharashtra's districts. This report maps and tests how demographic and work indicators are spread across the 35 districts of the 2011 Census. I mapped urbanisation, household size, sex ratio, total, male and female work participation, cultivator share and the Scheduled Caste and Scheduled Tribe shares in QGIS, correlated ten indicators in R, and used GeoDa for Moran's I, local clusters (LISA), OLS and a spatial error model.

The maps show substantial district-level variation. Urbanisation ranges from 11 to 100 per cent and the sex ratio from 832 to 1,122 females per 1,000 males. Female work participation varies much more widely across districts than male work participation, and total WPR is strongly correlated with female WPR (r = 0.94). The Global Moran's I results indicate positive spatial clustering for the seven indicators reported in the main table. In the regression analysis, urbanisation is negatively associated with total WPR across OLS, spatial lag and spatial error models. The spatial error model has the lowest AIC among the three specifications. These findings are descriptive and associational, not causal.

The results show strong regional structure that state-level figures can conceal. These are cross-sectional associations between districts, not causal effects. The spatial lag and spatial error specifications are reported in Section 11, while the weights definition and data limitations are documented in Sections 3, 8 and 14.

1. Introduction

Maharashtra is one of India's largest and most unevenly developed states. At the 2011 Census it held 112,374,333 people on 307,713 km², a density of 365 persons per km² (Census 2011 state profile). It has a metropolitan core around Mumbai, Thane and Pune, a forested and largely tribal north-west and east, a drier Deccan interior in Marathwada, and a cotton and orange belt in Vidarbha. For administration it is grouped into six divisions: Konkan, Pune, Nashik, Aurangabad, Amravati and Nagpur (List of districts of Maharashtra).

State averages hide most of this. The state's urban share was 45.2 per cent, its sex ratio 929 females per 1,000 males and its child sex ratio 894 (Census 2011 state profile). Across the 35 districts used here, the urban share runs from 11 to 100 per cent and the sex ratio from 832 to 1,122 (legends of the maps in Section 5). Total work participation, which averages 45.2 per cent across districts, runs from 39.9 to 54.5 per cent. A single state figure describes none of the districts at either end.

The question that guides this report is therefore a spatial one: do neighbouring districts resemble each other more than chance would suggest, and do the patterns of urbanisation, household size,

sex ratio and work line up with one another? An ordinary regression treats every district as an independent observation and cannot answer the first part. Spatial statistics were built for it

I follow the course workflow and report the steps in the order I worked through them, so that each map or output is interpreted where it first appears. I built and mapped the district layer in QGIS, described and correlated the data in R, and used GeoDa for spatial weights, Moran's I, local clusters and the spatial regressions. Section 2 sets out the objectives, Section 3 the data and methods, and Sections 4 to 11 the eight analytical steps. Section 12 pulls the findings together, Section 13 concludes and Section 14 states what the evidence cannot support.

Literacy is not mapped as a standalone indicator in the main QGIS map set. Total literacy is included as a control variable in the regression models.

2. Objectives

The objectives are limited to what the maps and outputs I have actually produced can support. The first two are descriptive and the last four are inferential.

1. To map the district-level distribution of urbanisation, average household size, sex ratio, total, male and female work participation, cultivators, and the Scheduled Caste (SC) and Scheduled Tribe (ST) population shares in Maharashtra, Census 2011.
2. To compare regions and genders in work participation, and to see how both relate to urbanisation.
3. To measure the correlation between the main indicators across the 35 districts.
4. To test, with Global Moran's I, whether urbanisation, household size, sex ratio, child sex ratio and the three work participation rates are spatially clustered.
5. To locate the clusters and outliers with Local Moran's I (LISA) for six of these indicators.
6. To model total work participation on urbanisation, household size and total literacy with OLS and a spatial error model, and to test whether the spatial error specification improves on OLS.

Cultivators, SC and ST shares are mapped only. I did not run spatial statistics on them, and I do not claim any.

3. Data and Methodology

3.1 Data and unit of analysis

The analysis uses district-level indicators attributed to Census of India 2011 for Maharashtra, joined to a district polygon layer named Maharashtra_District. The working dataset contains 35 district observations, and the maps carry the source note “Census of India, 2011”. The district is the unit of analysis throughout. GeoDa reports an expected Moran’s I value of −0.0294, consistent with n = 35. The exact original table identifier and the original download source for the boundary file were not retained in the working notes; this limits independent reproducibility and is acknowledged in Section 14. The official Census of India District Census

Handbooks provide district-level demographic and socioeconomic tables and are listed in the references.

3.2 The study area

In 2011 Maharashtra had 35 districts, with Mumbai City and Mumbai Suburban as separate districts. The state now has 36 because Palghar was separated from Thane after the Census (List of districts of Maharashtra), so every result here describes the 2011 map. For discussion I use the six administrative divisions and the common regional names (Konkan, western Maharashtra, Marathwada, Vidarbha). These are my own groupings, not Census variables. A district is a sensible unit because it is the finest level at which these Census indicators are published, but it averages over very different places inside it, as Section 14 notes.

3.3 Decoding the shapefile field names

The attribute fields are stored as MH Indic_4 to MH Indic_18. Each field was matched to an indicator using the project variable notes and checked against map legends and spatial patterns. Because a complete metadata sheet was not retained with the working files, some definitions should be treated as operational interpretations rather than independently verified metadata. The variable dictionary in Appendix A records how each field was used.

Field	Indicator	Used in this report as
MH Indic_4	Urbanisation rate (Urb_pct)	Mapped, correlated, Moran, LISA, regression
MH Indic_5	Average household size (HH_size)	Mapped, correlated, Moran, LISA, regression
MH Indic_6	Sex ratio	Mapped, correlated, Moran, LISA
MH Indic_7	Child sex ratio (0–6)	Correlated, Moran only (no map or LISA)
MH Indic_8	Total literacy rate	Regression control only
MH Indic11	Total work participation rate (Tot_WPR)	Mapped, correlated, Moran, LISA, regression (dependent)
MH Indic12	Male work participation rate	Mapped, correlated, Moran, LISA
MH Indic13	Female work participation rate	Mapped, correlated, Moran, LISA
MH Indic14	Cultivators (see 3.4)	Mapped
MH Indic17	Scheduled Caste population	Mapped

Field	Indicator	Used in this report as
	share	
MH Indic18	Scheduled Tribe population share	Mapped

The other literacy fields (Indic_9 and Indic10) and the male and female cultivator fields (Indic15 and Indic16) are not analysed.

3.4 Indicators

All indicators refer to district-level Census 2011 data. The formulas below describe the intended meaning of each indicator. Where the original metadata or denominator was not preserved, the report states this as a limitation rather than treating the definition as independently verified.

- **Urbanisation (%)** = urban population / total population × 100.
- **Average household size** = household population / number of households (persons per household).
- **Sex ratio** = females / males × 1,000. **Child sex ratio** = girls aged 0–6 / boys aged 0–6 × 1,000.
- Work participation rate (WPR, %) is interpreted as workers divided by the relevant population, multiplied by 100, for persons, males and females. The exact denominator in the source field was not independently verified from the original metadata. The maps and results are therefore interpreted as the supplied WPR indicators, and not as a newly calculated measure.
- Cultivator indicator. The map legend ranges from 0.4 to 48.3, indicating that the mapped field is a percentage or share rather than a count. The original denominator was not preserved in the project notes, so the report refers to it as the cultivator share and does not assign a more specific denominator. The map title “Total Cultivators” should be read as a label inherited from the layout, not evidence that the field is a head count.
- **SC and ST shares (%)** = SC (or ST) population / total population × 100.

A population share is not a measure of deprivation, and a low female WPR is not the same as an absence of women's economic contribution. Census work counts miss much unpaid and informal work, which matters for the interpretation of the work indicators.

3.5 The workflow at a glance

1. **QGIS:** build the 35-district layer, join the Census indicators and map them with five-class legends.
2. **R:** describe the data, map urbanisation in two ways, and compute correlations.
3. **GeoDa, spatial weights:** define neighbours for the 35 districts.
4. **GeoDa, autocorrelation:** compute Global Moran's I and Local Moran's I (LISA).

5. **GeoDa, regression:** estimate OLS and spatial models for total WPR.

The same 35-district layer is used at every step.

3.6 Methods in brief

Global Moran's I (Moran, 1950) measures whether similar values sit near each other across the whole map. Under spatial randomness its expected value is $-1/(n-1) = -0.0294$ here. Significance came from 999 random permutations in GeoDa (Anselin, Syabri and Kho, 2006), so the smallest possible pseudo p-value is 0.001. Local Moran's I (Anselin, 1995) splits the pattern into High-High and Low-Low clusters and High-Low and Low-High outliers. I mapped only districts with $p \leq 0.05$, which is the threshold shown on each LISA map, without adjusting for multiple tests. The regressions were estimated in GeoDa with the same weights; the spatial error model uses maximum likelihood

4. Step 1: Building the GIS Layer in QGIS

Everything starts from one layer, Maharashtra_District, which holds 35 district polygons with the Census indicators attached as the MH Indic fields described in Section 3.3. Each district keeps its name label on the maps, so the layer can be checked by eye against the administrative map.

Three features of the layer matter for what follows. First, the only map that states a coordinate reference system is the sex ratio map, which shows EPSG:4326. That is a geographic system in degrees, which is fine for a choropleth but means the kilometre scale bars on the maps are approximate. Second, a few labels are truncated ("Ahmadnaga", "Chand" for Chandrapur) and Osmanabad is unlabelled on some maps. Third, Mumbai City is too small to show as a filled polygon at state scale. Only a coastal sliver is visible, and I could not tell from the maps whether it is Mumbai City, Mumbai Suburban or both. This matters because the Mumbai districts are extreme observations for urbanisation, and a displaced or degenerate polygon would give them the wrong neighbours. The same layer was used in GeoDa, and Section 8 shows its connectivity graph.

5. Step 2: Mapping the Indicators in QGIS

Before any statistics I wanted to see the data. The nine maps below use five-class legends with unequal class widths, which is consistent with a Natural Breaks (Jenks) classification, although the maps do not state the method . The breaks therefore show relative position within Maharashtra and not absolute thresholds, and two different maps cannot be compared by colour alone, because the colour ramps also differ. District names below are read from the map labels and legends. I did not have the district attribute table, so I describe classes and not exact values. The first and last legend values are the minimum and maximum across the 35 districts.

Indicator	Minimum	Maximum	Legend breaks
Urbanisation (%)	11	100	11, 22, 33, 44, 77, 100

Indicator	Minimum	Maximum	Legend breaks
Household size (persons)	4.05	5.14	4.05, 4.2, 4.47, 4.72, 4.94, 5.14
Sex ratio (females per 1,000 males)	832	1,122	832, 886, 937, 966, 1,036, 1,122
Total WPR (%)	39.92	54.45	39.92, 40.86, 43.06, 45.66, 50.31, 54.45
Male WPR (%)	52.63	60.61	52.63, 53.48, 54.58, 56.08, 57.92, 60.61
Female WPR (%)	18.3	49.1	18.3, 24.7, 31.9, 37.9, 42.8, 49.1
Cultivators (%, see 3.4)	0.4	48.3	0.4, 11.1, 21.9, 31.7, 39.5, 48.3
Scheduled Caste (%)	2.9	20.1	2.9, 7.1, 11.3, 13.9, 16.7, 20.1
Scheduled Tribe (%)	0.6	69.3	0.6, 3.9, 9.5, 18.5, 38.7, 69.3

Table 1. Range and class breaks of the mapped indicators. Source: author's QGIS maps from Census 2011 data.

5.1 Urban structure and household size

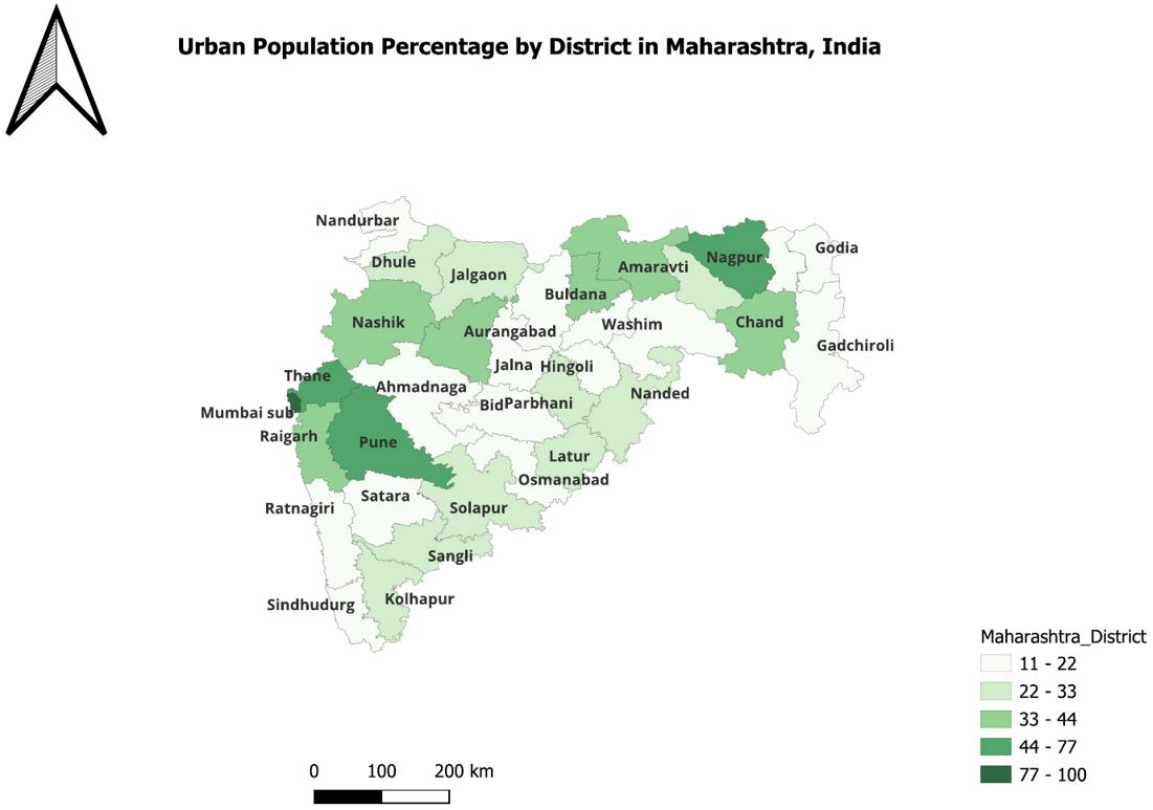


Figure 1. District-wise urban population (%), Maharashtra, Census 2011. Source: author's analysis using Census of India 2011 data in QGIS.

Reading Figure 1. The map shades each district by the share of its population that lives in urban areas. The legend has five classes (11–22, 22–33, 33–44, 44–77 and 77–100 per cent). Most districts fall in the two palest classes and only a few reach the darker ones, so the distribution is low with a short, steep upper tail. This agrees with the R histogram in Section 6.2.

District by district. The class of each district is as follows.

- **77–100 per cent:** Mumbai Suburban. Mumbai City is a coastal sliver that cannot be read separately at this map scale.
- **44–77 per cent:** Thane, Pune and Nagpur.
- **33–44 per cent:** Raigad, Nashik, Aurangabad, Buldana, Akola, Amravati and Chandrapur.
- **22–33 per cent:** Dhule, Jalgaon, Wardha, Nanded, Parbhani, Latur, Solapur, Sangli and Kolhapur.
- **11–22 per cent:** Nandurbar, Ahmednagar, Jalna, Bid, Hingoli, Osmanabad, Satara, Ratnagiri, Sindhudurg, Washim, Yavatmal, Bhandara, Gondia and Gadchiroli.

What the pattern shows. First, there is a metropolitan axis along the Konkan coast and Pune: Mumbai, Thane, Raigad and Pune all sit in the top three classes, and Nashik is in the third class beside them. Second, Nagpur is an inland urban node. It is the only Vidarbha district in the 44–

77 class, with Amravati, Akola, Buldana and Chandrapur one class below. Third, the lowest class forms large connected blocks: the south-west coast with Satara (Ratnagiri, Sindhudurg, Satara), the Marathwada interior with Ahmednagar (Jalna, Bid, Hingoli, Osmanabad, Ahmednagar), and eastern Vidarbha (Bhandara, Gondia, Gadchiroli). Nandurbar, Washim and Yavatmal are also in this class. This clumping is what the Moran's I of 0.56 for urbanisation measures in Section 9. In Section 10 it appears as a High-High cluster on the Thane–Raigad coast and Low-Low clusters at Gondia and Kolhapur. Kolhapur is in the 22–33 class, not the lowest. It is flagged because several of its neighbours (Sindhudurg, Ratnagiri and Satara) are in the lowest class.

Urbanisation is the variable that organises most of the others in this report. It has the strongest correlations with female WPR ($r = -0.82$), sex ratio ($r = -0.66$) and total WPR ($r = -0.65$). These are correlations across 35 districts and they do not show that urbanisation causes any of them.

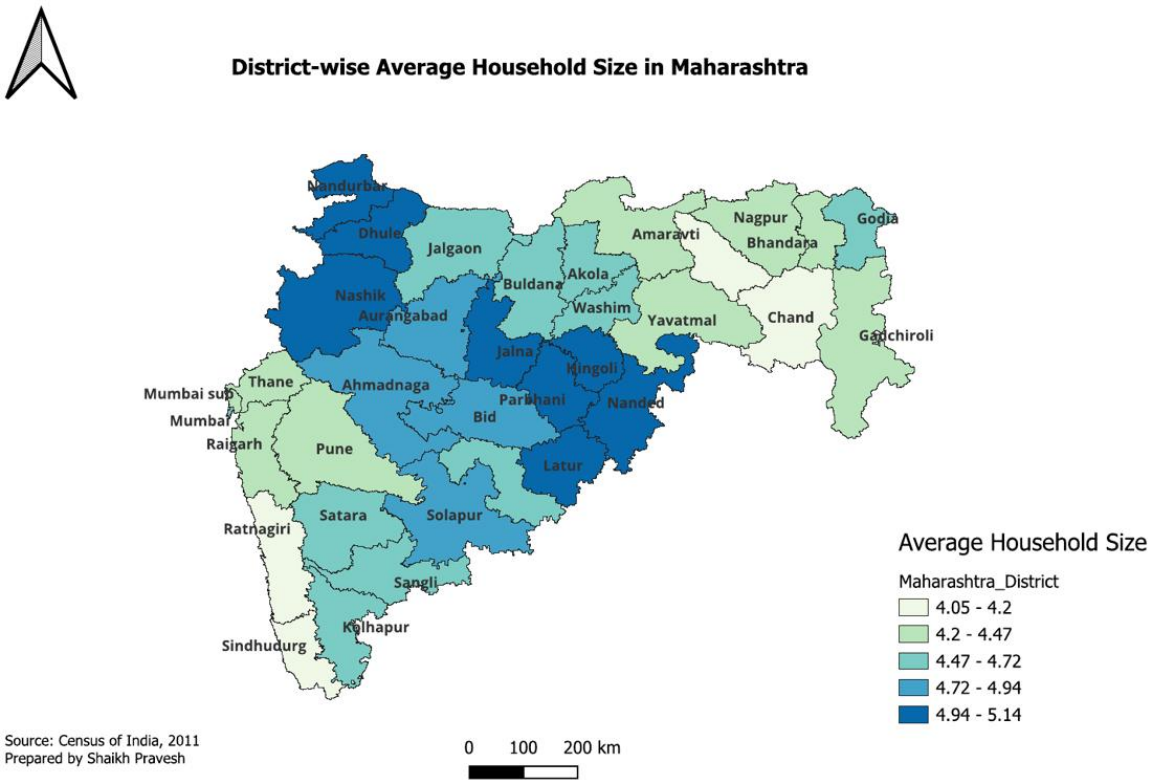


Figure 2. District-wise average household size, Maharashtra, Census 2011. Source: author's analysis using Census of India 2011 data in QGIS.

Reading Figure 2. Average household size runs from 4.05 to 5.14 persons. The five classes are 4.05–4.2, 4.2–4.47, 4.47–4.72, 4.72–4.94 and 4.94–5.14.

District by district.

- **4.94–5.14 (largest):** Nandurbar, Dhule, Nashik, Jalna, Hingoli, Parbhani, Nanded and Latur.
- **4.72–4.94:** Aurangabad, Ahmednagar, Bid and Solapur.

- **4.47–4.72:** Jalgaon, Buldana, Akola, Washim, Gondia, Osmanabad, Satara, Sangli and Kolhapur.
- **4.2–4.47:** Amravati, Yavatmal, Nagpur, Bhandara, Gadchiroli, Thane, Raigad and Pune.
- **4.05–4.2 (smallest):** Wardha, Chandrapur, Ratnagiri and Sindhudurg.

The Mumbai polygons are too small to read on this map.

What the pattern shows. The two largest classes make a largely connected belt of 12 districts. It runs from the north-west (Nandurbar, Dhule, Nashik) through Aurangabad, Ahmednagar, Jalna and Bid to Parbhani, Hingoli, Nanded and Latur, with Solapur at its southern end. The smallest households are on the southern Konkan coast (Ratnagiri, Sindhudurg) and in two Vidarbha districts (Wardha, Chandrapur). Household size is the most strongly clustered of the ten indicators (Moran's I = 0.61, Section 9). The LISA map in Section 10 marks nine High-High districts in the large-household belt and seven Low-Low districts, mostly in eastern Vidarbha.

The correlation matrix links larger households with lower male work participation (r = −0.62) and a lower child sex ratio (r = −0.52). The two maps show how this looks on the ground. Nandurbar, Aurangabad, Parbhani, Nanded and Latur are in the largest or second-largest household class and also in the lowest male WPR class (Figure 5). I read household size only as a descriptive demographic variable. It does not by itself indicate poverty, development or any behaviour.

5.2 Sex ratio

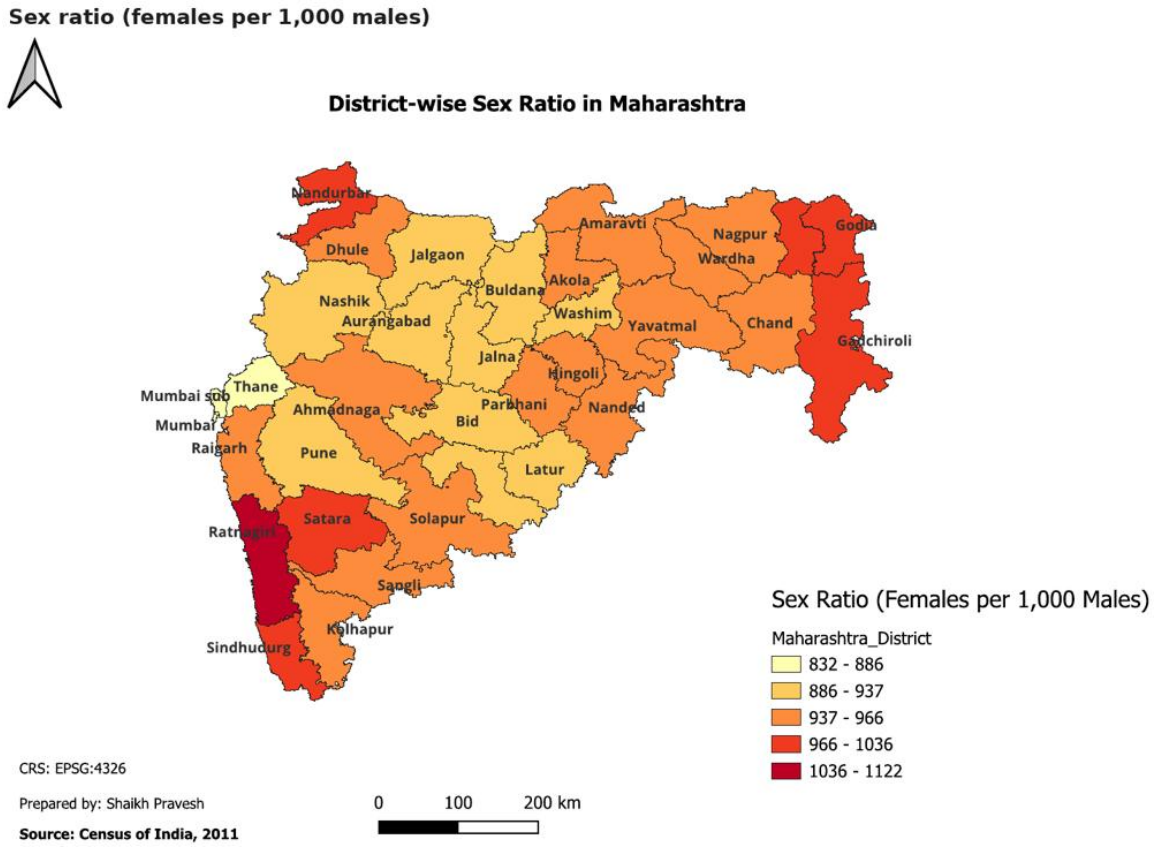


Figure 3. District-wise sex ratio (females per 1,000 males), Maharashtra, Census 2011. Source: author's analysis using Census of India 2011 data in QGIS. The child sex ratio was not mapped.

Reading Figure 3. The sex ratio, in females per 1,000 males, runs from 832 to 1,122. The five classes are 832–886, 886–937, 937–966, 966–1,036 and 1,036–1,122.

District by district.

- **1,036–1,122 (highest):** Ratnagiri.
- **966–1,036:** Satara, Sindhudurg, Nandurbar, Bhandara, Gondia and Gadchiroli.
- **937–966:** Raigad, Dhule, Ahmednagar, Akola, Amravati, Wardha, Nagpur, Chandrapur, Yavatmal, Hingoli, Parbhani, Nanded, Solapur, Sangli and Kolhapur.
- **886–937:** Pune, Nashik, Jalgaon, Aurangabad, Jalna, Buldana, Washim, Bid, Osmanabad and Latur.
- **832–886 (lowest):** Thane and Mumbai Suburban. Mumbai City cannot be read separately.

What the pattern shows. There are three bands. A high band runs down the south-west (Ratnagiri, Sindhudurg, Satara) and appears again on the north-western and north-eastern margins (Nandurbar, Bhandara, Gondia, Gadchiroli). A broad low band, one class above the minimum, crosses the central plateau from Nashik through Aurangabad, Jalna and Bid to Osmanabad and Latur, and also takes in Pune, Jalgaon, Buldana and Washim. The lowest class is the Thane–Mumbai coast. Section 10 turns this into three clusters: High-High in Ratnagiri, Sindhudurg, Kolhapur and Sangli, and Low-Low in Ahmednagar and the two Mumbai districts. Ahmednagar is in the 937–966 class itself. It is flagged because most of its neighbours (Nashik, Aurangabad, Bid, Osmanabad, Pune and Thane) are in a lower class.

The map is not simply 'urban means low'. Nagpur is in the 44–77 per cent urban class but is in the 937–966 sex ratio class, and Raigad is also in that class. The correlation of −0.66 with urbanisation is strong but far from perfect, and the map shows why. A skewed sex ratio in cities is often linked to male in-migration, but I have no migration data, so that stays a hypothesis.

5.3 Work participation

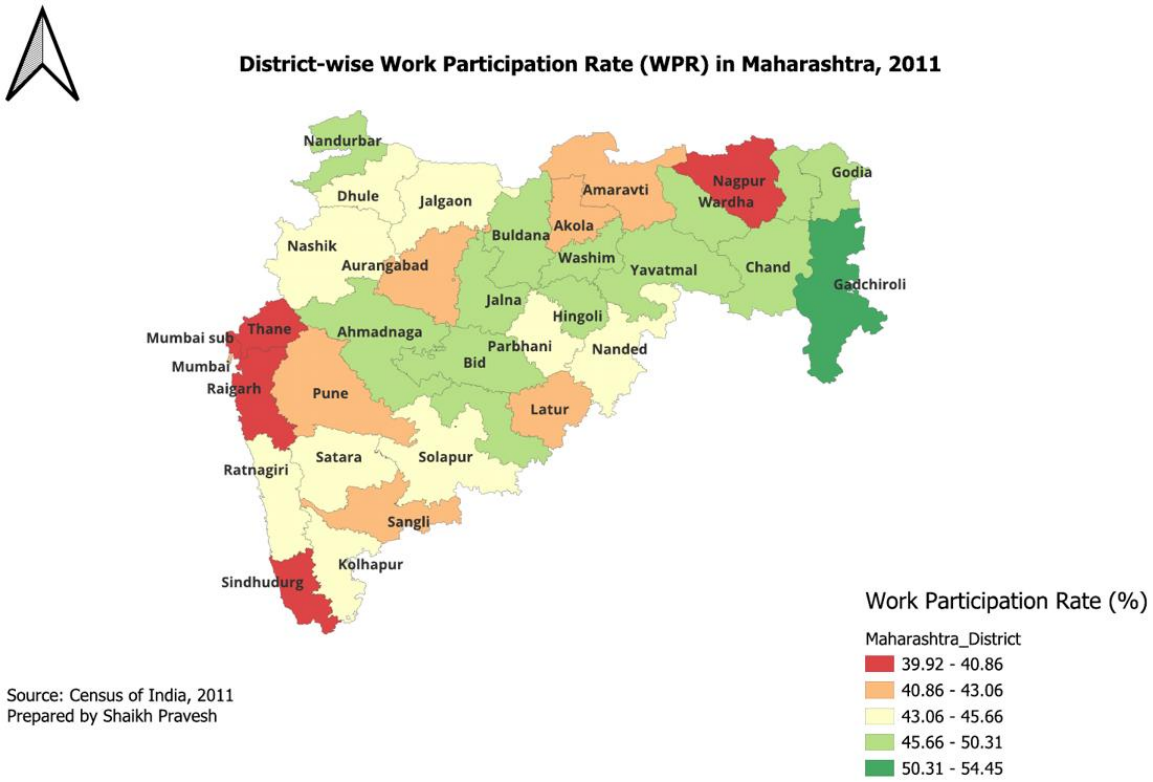


Figure 4. District-wise total work participation rate (WPR, %), Maharashtra, Census 2011. Source: author's analysis using Census of India 2011 data in QGIS.

Reading Figure 4. Total work participation rate (WPR) runs from 39.92 to 54.45 per cent. The classes are 39.92–40.86, 40.86–43.06, 43.06–45.66, 45.66–50.31 and 50.31–54.45.

District by district.

- **50.31–54.45 (highest):** Gadchiroli.
- **45.66–50.31:** Nandurbar, Buldana, Washim, Yavatmal, Wardha, Bhandara, Gondia, Chandrapur, Jalna, Hingoli, Bid, Ahmednagar and Osmanabad.
- **43.06–45.66:** Dhule, Jalgaon, Nashik, Parbhani, Nanded, Solapur, Satara, Kolhapur and Ratnagiri.
- **40.86–43.06:** Amravati, Akola, Aurangabad, Latur, Pune and Sangli.
- **39.92–40.86 (lowest):** Nagpur, Thane, Mumbai Suburban, Raigad and Sindhudurg. Mumbai City cannot be read separately.

What the pattern shows. Eastern and central Vidarbha (Wardha, Bhandara, Gondia, Chandrapur, Yavatmal, Washim, Buldana) and the Marathwada–Ahmednagar belt (Jalna, Bid, Hingoli, Osmanabad, Ahmednagar) are mostly in the second-highest class, with Gadchiroli alone above them. The low classes are scattered rather than connected. They include the metropolitan coast (Thane, Mumbai, Raigad), Sindhudurg, the inland city district of Nagpur, and Pune, Sangli,

Latur, Aurangabad, Akola and Amravati. Western and northern Maharashtra (Dhule, Jalgaon, Nashik) and the southern districts (Satara, Solapur, Kolhapur, Ratnagiri) share the middle class.

Section 10 finds only two clear regional clusters for this indicator: High-High in Bhandara, Gondia, Gadchiroli and Chandrapur, and Low-Low in Mumbai Suburban, Mumbai City, Raigad and Ratnagiri. Thane is in the lowest class but is not flagged, and Ratnagiri is flagged although it is in the middle class, so the cluster map reflects a district's neighbours as well as its own rank. Moran's I for total WPR (0.43) is the lowest of the ten indicators, which fits a map whose low classes are scattered.

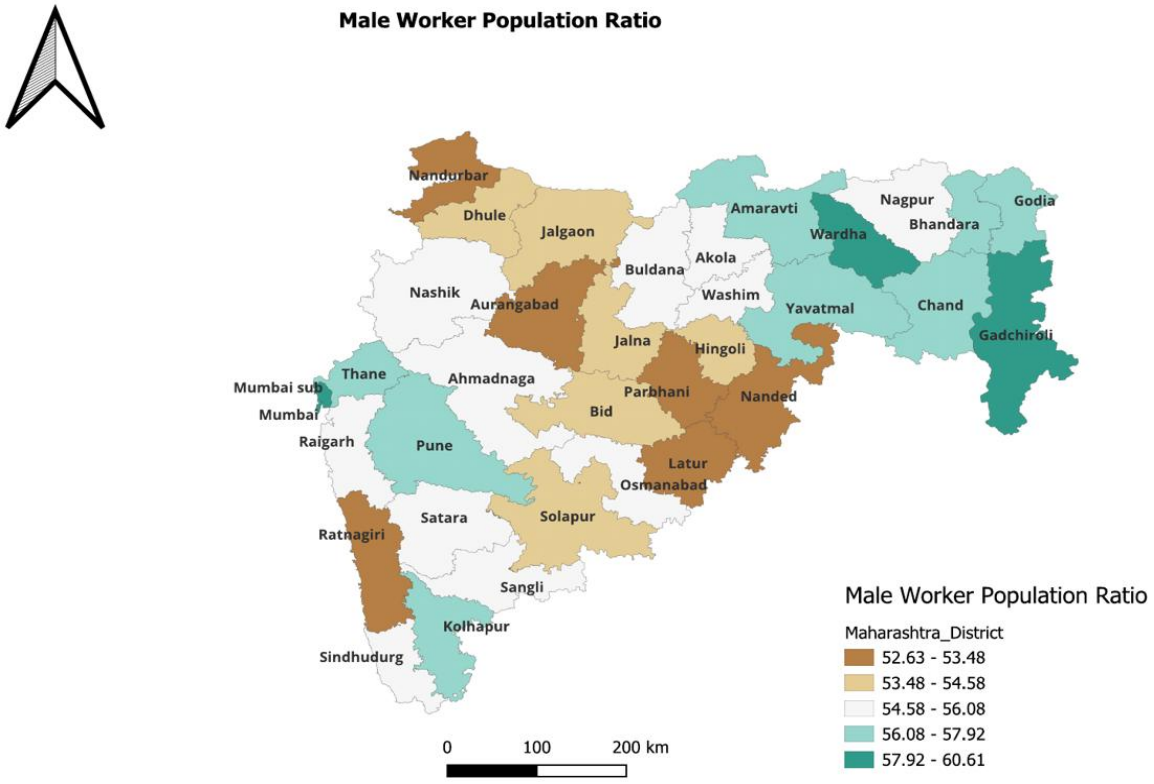


Figure 5. District-wise male work participation rate (WPR, %), Maharashtra, Census 2011. Source: author's analysis using Census of India 2011 data in QGIS.

The male work-participation map layout uses the label "Male Worker Population Ratio", while the rest of the report uses work participation rate (WPR). Since the original field metadata was not retained, the report analyses the supplied indicator as male WPR and notes this naming limitation rather than assuming the labels are perfectly interchangeable.

District by district.

- **57.92–60.61 (highest):** Wardha, Gadchiroli and Mumbai Suburban.
- **56.08–57.92:** Amravati, Yavatmal, Chandrapur, Gondia, Thane, Pune and Kolhapur.
- **54.58–56.08:** Nashik, Ahmednagar, Buldana, Akola, Washim, Nagpur, Bhandara, Osmanabad, Satara, Sangli, Raigad and Sindhudurg.
- **53.48–54.58:** Dhule, Jalgaon, Jalna, Hingoli, Bid and Solapur.

- **52.63–53.48 (lowest):** Nandurbar, Aurangabad, Parbhani, Nanded, Latur and Ratnagiri.

Mumbai City cannot be read separately.

What the pattern shows. The lowest male classes are concentrated in Marathwada (Aurangabad, Parbhani, Nanded and Latur, with Jalna, Hingoli and Bid one class above), plus Nandurbar and Ratnagiri. The highest values are in eastern Vidarbha (Wardha and Gadchiroli, with Amravati, Yavatmal, Chandrapur and Gondia in the next class) and in the western metropolitan region (Mumbai Suburban, Thane, Pune). The Marathwada low block is the Low-Low LISA cluster of Section 10, and the eastern Vidarbha block is the High-High cluster.

Compare this with Figure 4. Thane, Raigad and Nagpur are in the lowest total WPR class, but Thane is in the 56.08–57.92 male class and Raigad and Nagpur are in the 54.58–56.08 class. Their low total participation therefore comes from women's participation. This is what the correlations imply (total with female r = 0.94, total with male r = 0.09).

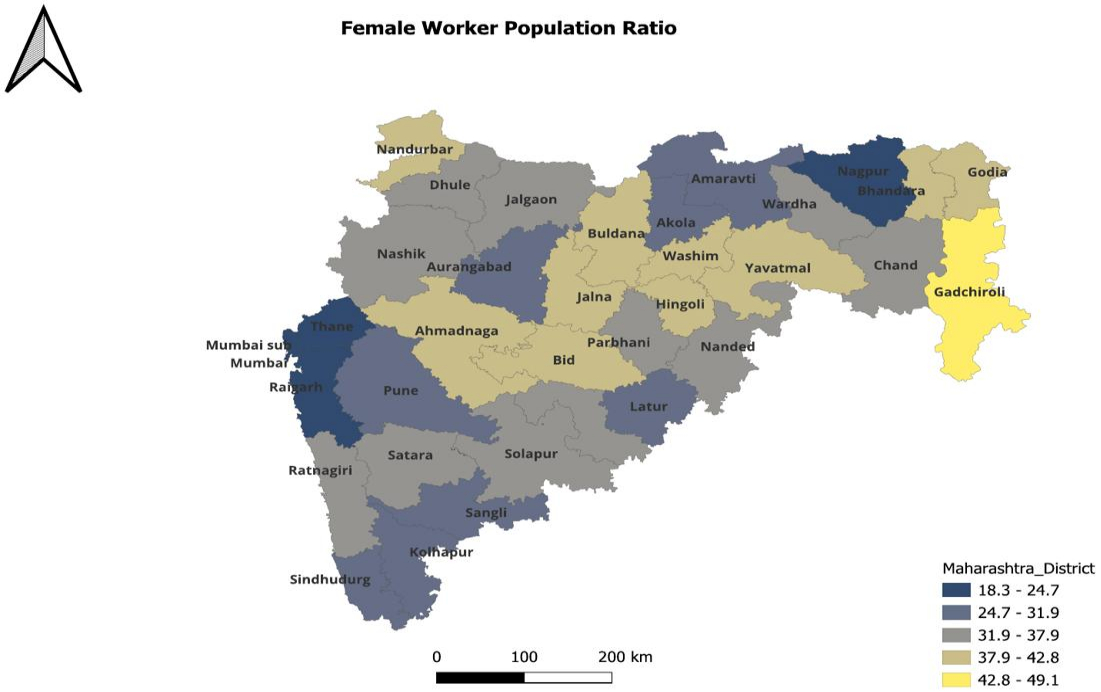


Figure 6. District-wise female work participation rate (WPR, %), Maharashtra, Census 2011. Source: author's analysis using Census of India 2011 data in QGIS.

Reading Figure 6. Female WPR varies far more than male WPR, from 18.3 to 49.1 per cent. The classes are 18.3–24.7, 24.7–31.9, 31.9–37.9, 37.9–42.8 and 42.8–49.1.

District by district.

- **42.8–49.1 (highest):** Gadchiroli.
- **37.9–42.8:** Nandurbar, Buldana, Washim, Yavatmal, Bhandara, Gondia, Jalna, Hingoli, Bid and Ahmednagar.

- **31.9–37.9:** Dhule, Jalgaon, Nashik, Wardha, Chandrapur, Parbhani, Nanded, Osmanabad, Solapur, Satara and Ratnagiri.
- **24.7–31.9:** Amravati, Akola, Aurangabad, Latur, Pune, Sangli, Kolhapur and Sindhudurg.
- **18.3–24.7 (lowest):** Nagpur, Thane, Raigad and the two Mumbai districts.

What the pattern shows. This is the map with the sharpest regional contrast. The lowest class is the metropolitan coast (Thane, Mumbai, Raigad) and Nagpur, all of which are in the upper urbanisation classes, and the correlation with urbanisation is $r = -0.82$. The highest class is Gadchiroli alone. A crescent of the second-highest class runs from Nandurbar in the north-west through Buldana, Washim, Yavatmal, Bhandara and Gondia to Jalna, Hingoli, Bid and Ahmednagar in the centre. There are exceptions that stop the relation from being mechanical. Pune is in the 44–77 per cent urban class but only in the second-lowest female class, and Sindhudurg, one of the least urban districts, is also in the second-lowest class.

Section 10 records Gondia, Gadchiroli and Hingoli as High-High and the Thane–Mumbai–Raigad coast as Low-Low. Aurangabad is the single Low-High district: its own female WPR is in a low class while most of its neighbours (Jalna, Ahmednagar, Bid and Buldana) are in the 37.9–42.8 class.

The male range is about 8 points and the female range about 31 points. The map of total WPR therefore looks like the female map, and Section 7 confirms it ($r = 0.94$ between total and female WPR, but only 0.09 between total and male). Women's work participation is what separates the districts. Because the Census counts only recorded work, a low female WPR in a metropolitan district does not mean low economic activity. It more likely reflects what is counted and where women work.

5.4 Cultivators, Scheduled Castes and Scheduled Tribes

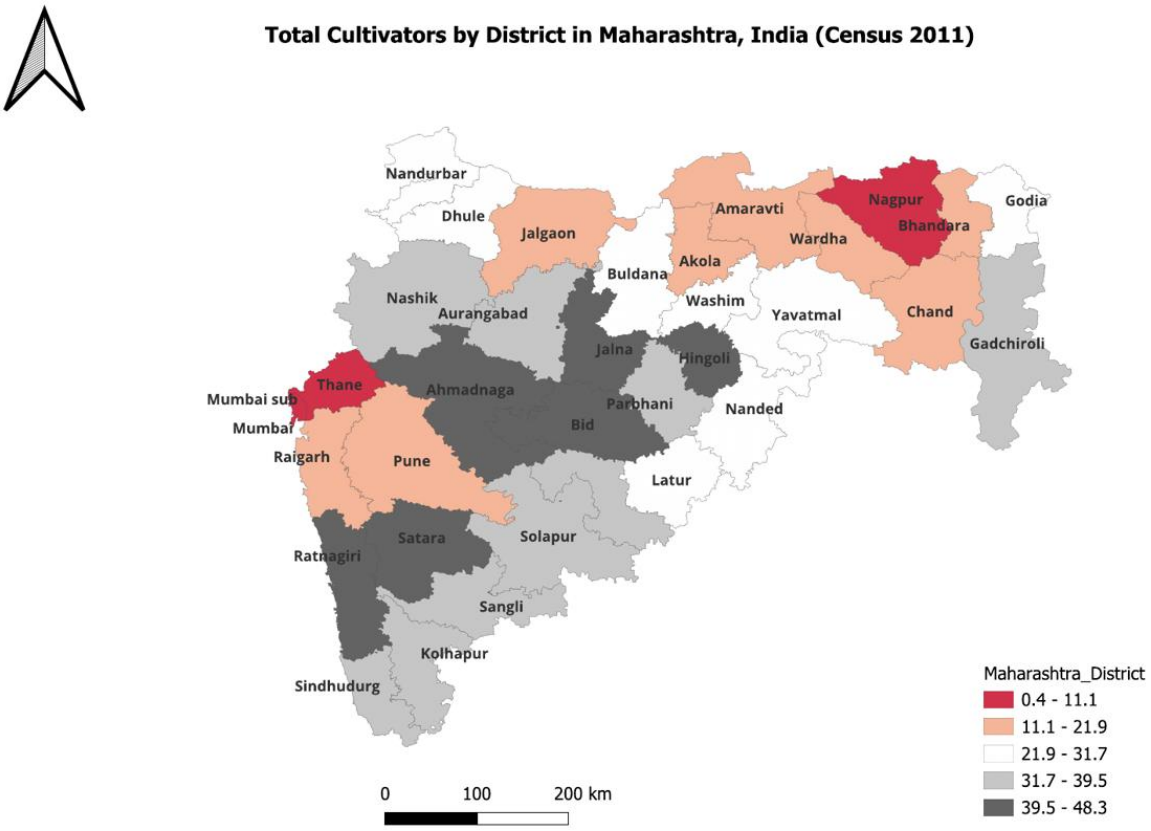


Figure 7. District-wise cultivator share (see Section 3.4), Maharashtra, Census 2011. Source: author's analysis using Census of India 2011 data in QGIS.

Reading Figure 7. The layout title says 'Total Cultivators', but the legend runs from 0.4 to 48.3 and so is a share and not a count (the denominator is unconfirmed, see Section 3.4). The classes are 0.4–11.1, 11.1–21.9, 21.9–31.7, 31.7–39.5 and 39.5–48.3.

District by district.

- **39.5–48.3 (highest):** Ahmednagar, Jalna, Bid, Hingoli, Satara and Ratnagiri.
- **31.7–39.5:** Nashik, Aurangabad, Parbhani, Gadchiroli, Solapur, Osmanabad, Sangli, Kolhapur and Sindhudurg.
- **21.9–31.7:** Nandurbar, Dhule, Buldana, Washim, Yavatmal, Gondia, Nanded and Latur.
- **11.1–21.9:** Jalgaon, Akola, Amravati, Wardha, Bhandara, Chandrapur, Raigad and Pune.
- **0.4–11.1 (lowest):** Nagpur, Thane and Mumbai Suburban.

What the pattern shows. The highest class is a connected band across the centre (Ahmednagar, Jalna, Bid, Hingoli) plus a south-western strip (Satara, Ratnagiri). The lowest class is the three most urban districts, Thane, Mumbai Suburban and Nagpur. Thane and Nagpur are also in the lowest total and female WPR classes (Figures 4 and 6). This is a share of workers who are cultivators, so it does not measure agricultural dependence, because agricultural labourers are not included and the denominator is unconfirmed. I did not compute correlations for this variable.

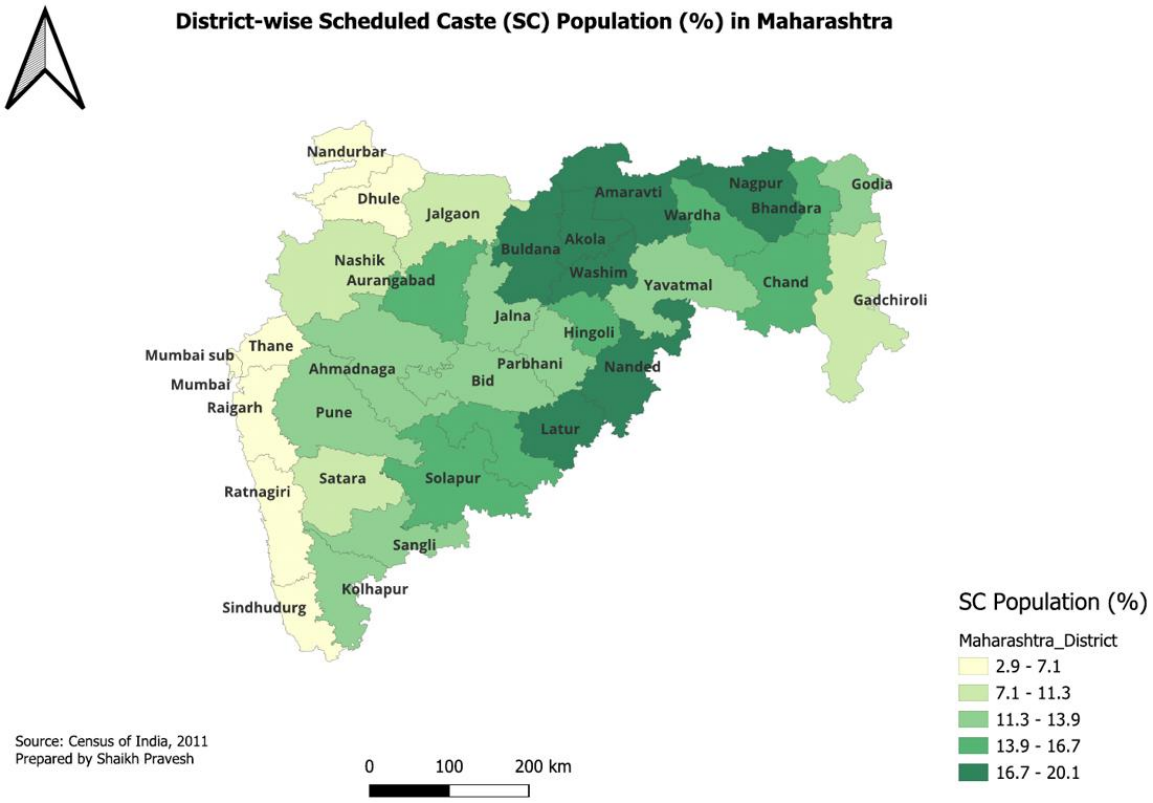


Figure 8. District-wise Scheduled Caste (SC) population share (%), Maharashtra, Census 2011. Source: author's analysis using Census of India 2011 data in QGIS.

Reading Figure 8. The Scheduled Caste share runs from 2.9 to 20.1 per cent. The classes are 2.9–7.1, 7.1–11.3, 11.3–13.9, 13.9–16.7 and 16.7–20.1.

District by district.

- **16.7–20.1 (highest):** Amravati, Akola, Buldana, Washim, Nagpur, Nanded and Latur.
- **13.9–16.7:** Wardha, Bhandara, Chandrapur, Aurangabad, Hingoli, Solapur and Osmanabad.
- **11.3–13.9:** Gondia, Yavatmal, Jalna, Parbhani, Bid, Ahmednagar, Pune, Sangli and Kolhapur.
- **7.1–11.3:** Jalgaon, Nashik, Satara and Gadchiroli.
- **2.9–7.1 (lowest):** Nandurbar, Dhule, Thane, Raigad, Ratnagiri and Sindhudurg.

The two Mumbai polygons are too small to read.

What the pattern shows. The highest classes form one block in north-central and eastern Vidarbha (Amravati, Akola, Buldana, Washim, Nagpur, Wardha, Bhandara, Chandrapur) that continues south-west through Hingoli and Nanded to Latur, Osmanabad and Solapur. The lowest class is the north-western edge (Nandurbar, Dhule) and the Konkan coast from Thane to Sindhudurg. In the five-variable correlation matrix in the R output, the SC share has no correlation stronger than 0.21 in absolute terms with household size, urbanisation or work

participation. The map therefore describes where the Scheduled Caste population is concentrated and does not link that to work participation.

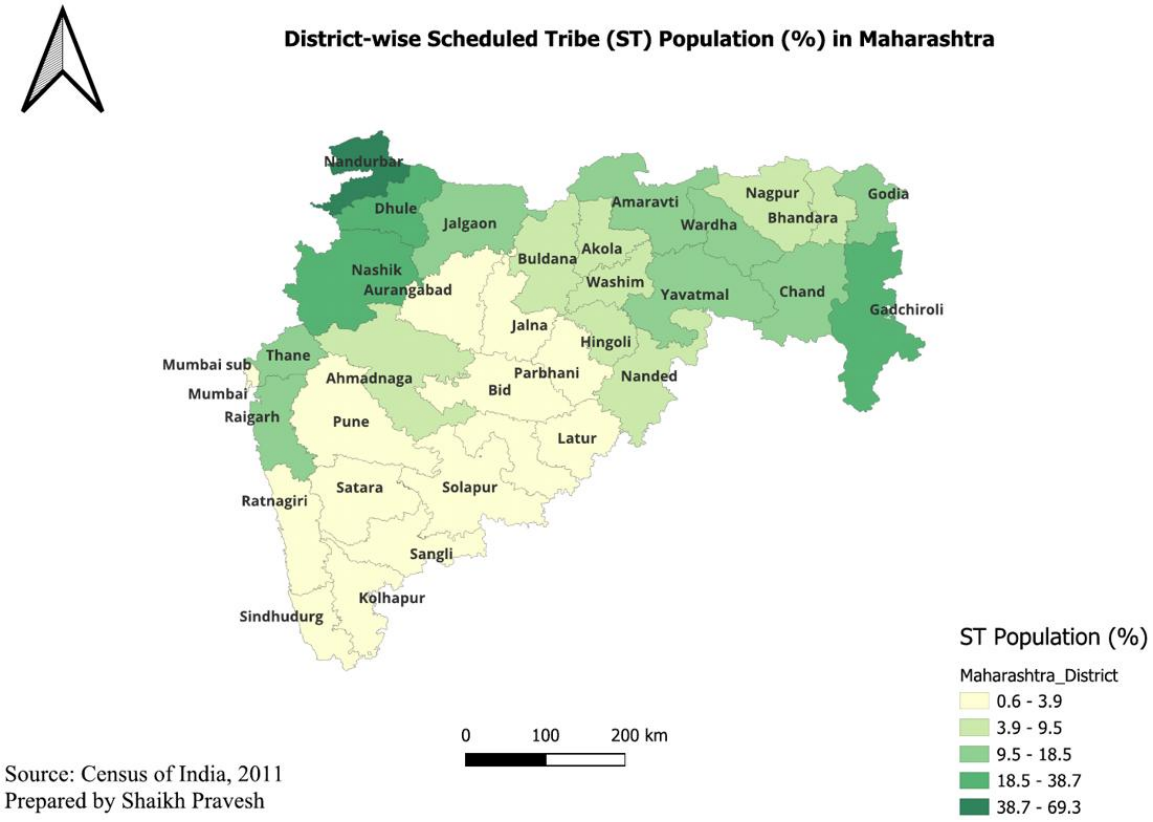


Figure 9. District-wise Scheduled Tribe (ST) population share (%), Maharashtra, Census 2011. Source: author's analysis using Census of India 2011 data in QGIS.

Reading Figure 9. The Scheduled Tribe share is the most unequal variable on the maps, from 0.6 to 69.3 per cent, so the class breaks are very uneven: 0.6–3.9, 3.9–9.5, 9.5–18.5, 18.5–38.7 and 38.7–69.3.

District by district.

- **38.7–69.3 (highest):** Nandurbar.
- **18.5–38.7:** Dhule, Nashik and Gadchiroli.
- **9.5–18.5:** Jalgaon, Thane, Raigad, Amravati, Wardha, Yavatmal, Chandrapur and Gondia.
- **3.9–9.5:** Ahmednagar, Buldana, Akola, Washim, Hingoli, Nanded, Nagpur and Bhandara.
- **0.6–3.9 (lowest):** Aurangabad, Jalna, Parbhani, Bid, Latur, Osmanabad, Pune, Satara, Solapur, Sangli, Kolhapur, Ratnagiri and Sindhudurg.

The two Mumbai polygons are too small to read.

What the pattern shows. There are two tribal belts. One lies along the northern Sahyadri and the north-west (Nandurbar, Dhule, Nashik, with Jalgaon, Thane and Raigad in the 9.5–18.5

class). The other is in the east (Gadchiroli, with Chandrapur, Gondia, Yavatmal, Wardha and Amravati in the 9.5–18.5 class). A large block of 13 districts across Marathwada and western and southern Maharashtra is below 3.9 per cent. The ST and SC maps are largely complementary. Nandurbar and Dhule are in the lowest SC class, and the highest SC class lies almost entirely in districts with an ST share below 9.5 per cent, with Amravati the one exception.

Gadchiroli is where several maps coincide: ST share in the 18.5–38.7 class, the highest total and female WPR classes, the lowest urbanisation class and the second-lowest SC class. These coincidences are descriptive. I did not compute correlations for the ST share, so the maps do not support any claim about how tribal share and work participation are related.

5.5 All 35 districts on the nine maps

Table 1a puts the nine maps side by side. Each cell is the legend class of the district on that map, from 1 (the lowest legend class) to 5 (the highest). Class 5 therefore means the highest urban share, the largest households, the highest sex ratio, the highest participation, the highest cultivator share and so on, and the class ranges are those in Table 1. 'n/r' means the district is too small to read on that map. The table lets the reader compare a district across indicators without going back to nine figures. It records classes read from the maps and not the underlying values.

Division	District	Urban	HH size	Sex ratio	Total WPR	Male WPR	Female WPR	Cultiv.	SC	ST
Konkan	Thane	4	2	1	1	4	1	1	1	3
Konkan	Mumbai Suburban	5	n/r	1	1	5	1	1	n/r	n/r
Konkan	Mumbai City	n/r	n/r	n/r	n/r	n/r	1	n/r	n/r	n/r
Konkan	Raigad	3	2	3	1	3	1	2	1	3
Konkan	Ratnagiri	1	1	5	3	1	3	5	1	1
Konkan	Sindhudurg	1	1	4	1	3	2	4	1	1
Nashik	Nandurbar	1	5	4	4	1	4	3	1	5
Nashik	Dhule	2	5	3	3	2	3	3	1	4

Division	District	Urban	HH size	Sex ratio	Total WPR	Male WPR	Female WPR	Cultiv.	SC	ST
Nashik	Jalgaon	2	3	2	3	2	3	2	2	3
Nashik	Nashik	3	5	2	3	3	3	4	2	4
Nashik	Ahmednagar	1	4	3	4	3	4	5	3	2
Pune	Pune	4	2	2	2	4	2	2	3	1
Pune	Satara	1	3	4	3	3	3	5	2	1
Pune	Sangli	2	3	3	2	3	2	4	3	1
Pune	Solapur	2	4	3	3	2	3	4	4	1
Pune	Kolhapur	2	3	3	3	4	2	4	3	1
Aurangabad	Aurangabad	3	4	2	2	1	2	4	4	1
Aurangabad	Jalna	1	5	2	4	2	4	5	3	1
Aurangabad	Parbhani	2	5	3	3	1	3	4	3	1
Aurangabad	Hingoli	1	5	3	4	2	4	5	4	2
Aurangabad	Nanded	2	5	3	3	1	3	3	5	2
Aurangabad	Latur	2	5	2	2	1	2	3	5	1
Aurangabad	Osmanabad	1	3	2	4	3	3	4	4	1
Aurangabad	Bid	1	4	2	4	2	4	5	3	1
Amravati	Buldana	3	3	2	4	3	4	3	5	2

Division	District	Urban	HH size	Sex ratio	Total WPR	Male WPR	Female WPR	Cultiv.	SC	ST
Amravati	Akola	3	3	3	2	3	2	2	5	2
Amravati	Washim	1	3	2	4	3	4	3	5	2
Amravati	Amravati	3	2	3	2	4	2	2	5	3
Amravati	Yavatmal	1	2	3	4	4	4	3	3	3
Nagpur	Nagpur	4	2	3	1	3	1	1	5	2
Nagpur	Wardha	2	1	3	4	5	3	2	4	3
Nagpur	Bhandara	1	2	4	4	3	4	2	4	2
Nagpur	Gondia	1	3	4	4	4	4	3	3	3
Nagpur	Chandrapur	3	1	3	4	4	3	2	4	3
Nagpur	Gadchiroli	1	2	4	5	5	5	4	2	4

Table 1a. Legend class (1 = lowest, 5 = highest) of each district on the nine QGIS maps. Mumbai City is a sliver and can be read only on the female WPR map. Source: author's reading of the QGIS maps in Figures 1 to 9. Division membership follows the 2011 administrative divisions.

5.6 What the nine maps say together

Reading across the columns of Table 1a gives a regional picture that no single map shows.

- **Konkan.** The coast splits in two. Thane, Mumbai and Raigad are the urban, low-participation end: Thane, Mumbai Suburban and Raigad are in the lowest total WPR class, all of them are in the lowest female WPR class, Thane and Mumbai are in the lowest sex ratio class, and Thane and Raigad are in the lowest SC class. Ratnagiri and Sindhudurg are the opposite: in the lowest urbanisation and household-size classes, with

the highest and second-highest sex ratios and high cultivator shares, although Sindhudurg is also in the lowest total WPR class. The Konkan is therefore not one regime.

- **Nashik division.** Nandurbar, Dhule and Nashik share the largest households, the highest ST shares and the lowest or low SC shares. Ahmednagar is different: it has the highest cultivator class, the lowest urbanisation class and the second-highest female WPR class, and it is a Low-Low sex ratio cluster in Section 10.
- **Pune division.** Pune district is a metropolitan node with high male participation but low female participation. Satara, Sangli, Solapur and Kolhapur are all in the lowest ST class and in the low or middle urbanisation classes. Sangli and Kolhapur, with Ratnagiri and Sindhudurg, form the High-High sex ratio cluster. Satara also has a high sex ratio but is not flagged.
- **Aurangabad (Marathwada).** This division has the clearest multi-map signature: large households, the lowest male participation class (Aurangabad, Parbhani, Nanded, Latur), the highest cultivator class (Jalna, Bid, Hingoli), very low ST shares and mostly low urbanisation. Nanded and Latur also have the highest SC class.
- **Amravati division.** Buldana, Akola, Washim and Amravati are in the highest SC class. Their other classes are mostly middle ones, with a sex ratio in the 886–966 range and low-to-middle urbanisation.
- **Nagpur division.** Nagpur district is the anomaly: it is in the 44–77 urban class and in the lowest classes for total WPR, female WPR and cultivator share, yet in the highest class for SC. Its neighbours Bhandara, Gondia and Gadchiroli are in the lowest urbanisation class and the second-highest sex ratio class. Bhandara, Gondia, Chandrapur and Gadchiroli are in the top two total WPR classes, and Gadchiroli is the only district in the top class for total, male and female WPR together.

These are descriptive comparisons across nine maps of the same 35 districts. They do not show why the regions differ, and the data here cannot separate the effect of any one variable from the others

6. Step 3: Descriptive Statistics and Urbanisation in R

In R I looked at distributions, tried log transformations, and drew the urbanisation data in two ways. The only summary statistics available to me are for total WPR, taken from the regression output in Section 11: across the 35 districts its mean is 45.21 per cent and its standard deviation 3.31, with a range of 39.92 to 54.45 per cent. The district-level table behind the other indicators was not supplied, so I do not report means and standard deviations for them.

6.1 Urbanisation in R: polygons and a grid

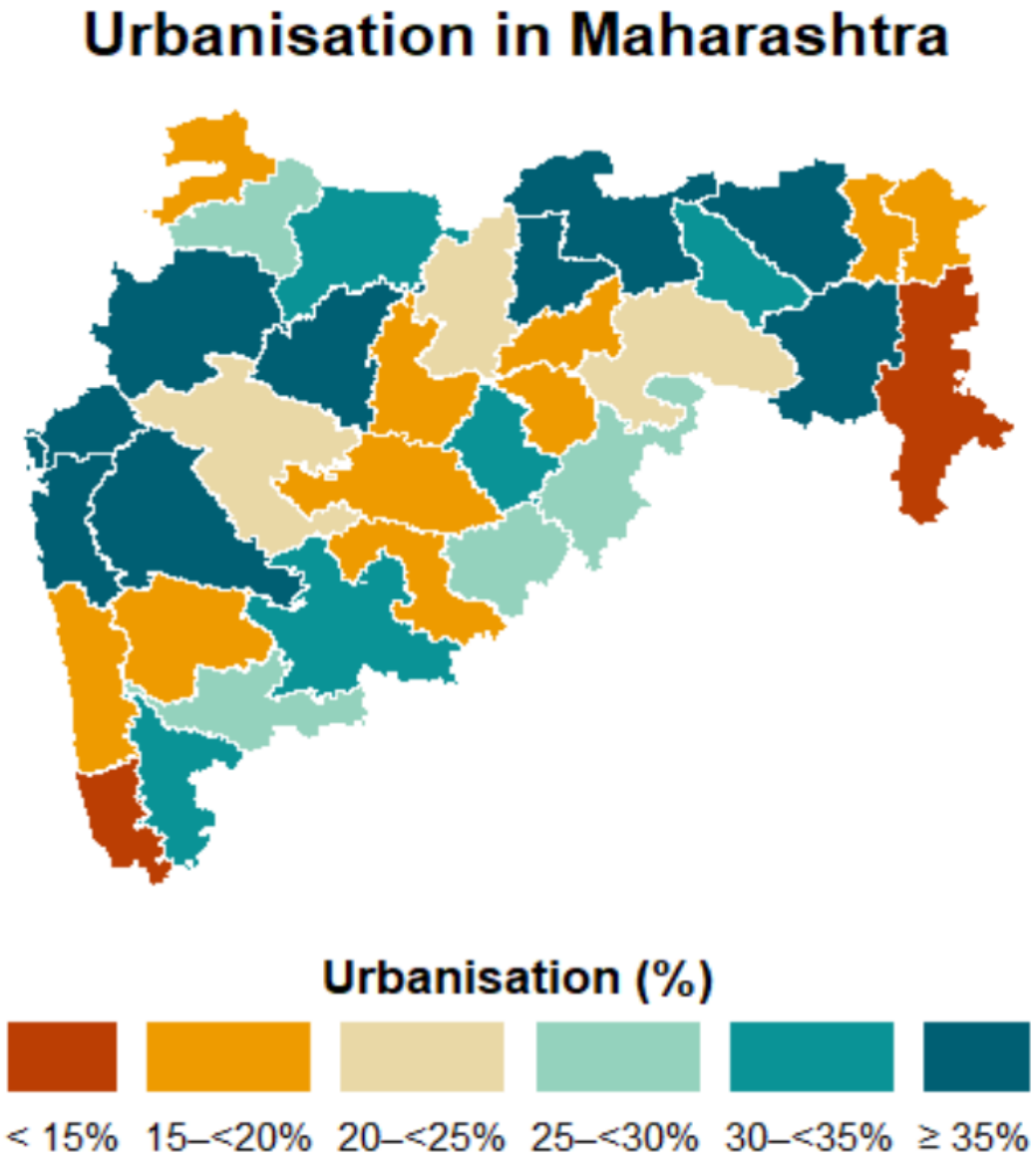


Figure 10. Urbanisation classes across the districts, polygon choropleth in R. Source: author's analysis using Census of India 2011 data.

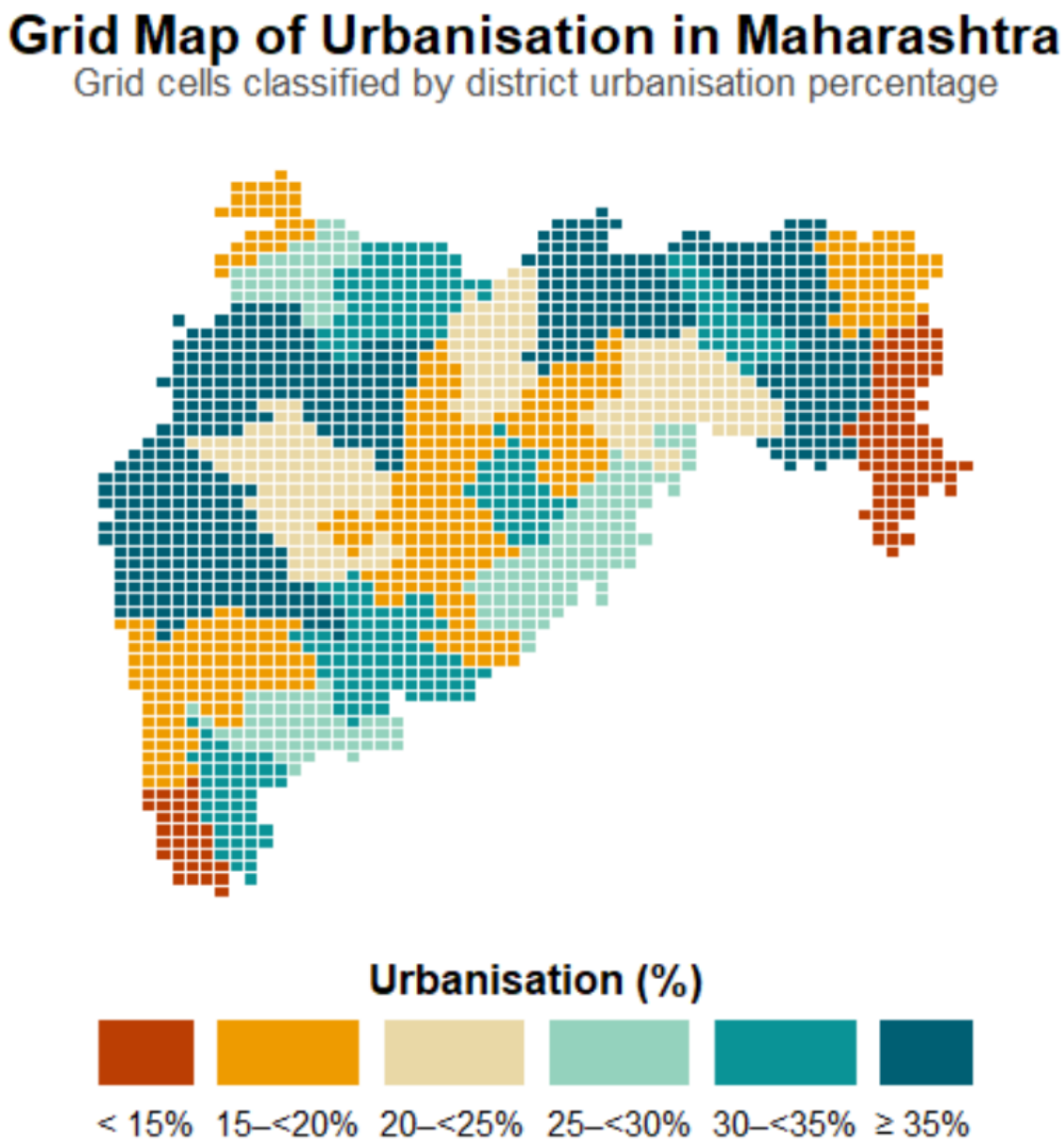


Figure 11. Urbanisation classes across the districts, grid map in R. Source: author's analysis using Census of India 2011 data.

The R maps use six classes (below 15, 15–20, 20–25, 25–30, 30–35 and 35 per cent or more), which is finer than the QGIS legend at the low end, where most districts fall. The grid map (Figure 11) turns each district into equal-sized grid cells coloured by its class. This stops large districts such as Gadchiroli from dominating the picture and gives a more even view of how many districts sit in each class. Gadchiroli and Sindhudurg are the districts shown below 15 per cent. Read against Figure 1, the two R maps show the same structure with a finer low end. The darkest class (35 per cent or more) is the western block around Nashik, Pune and the Konkan coast, plus the Nagpur area. The orange districts (15 to under 20 per cent) include the Ahmednagar–Bid–Jalna belt, the north-western tip at Nandurbar, the Gondia–Bhandara area and

the south-west coast around Ratnagiri. These are the districts that QGIS places in its lowest classes. Because two separate pieces of software give the same pattern, I am more confident that the class reading in Section 5 is correct, although I could not check exact values.

6.2 Distribution and household size against urbanisation

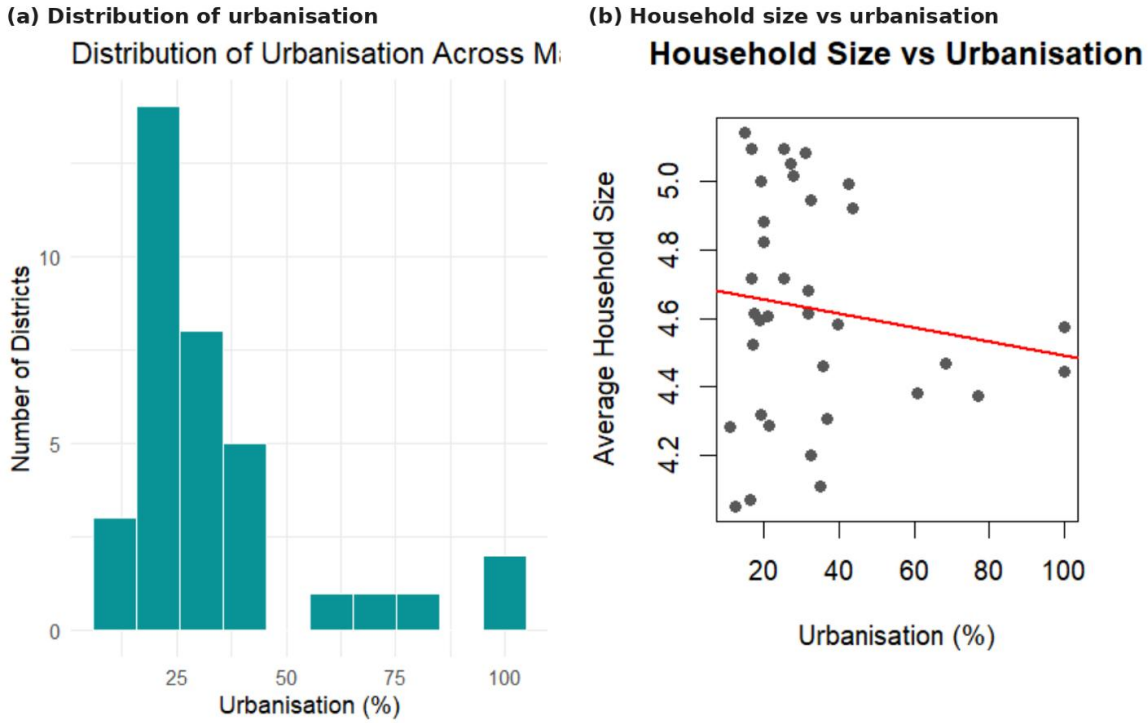
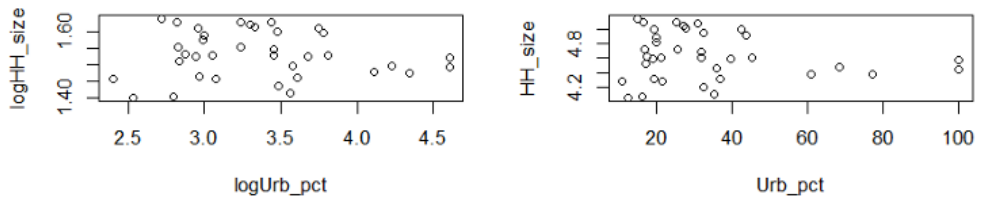


Figure 12. (a) Distribution of urbanisation across the 35 districts; (b) average household size against urbanisation, with the fitted line. Source: author's analysis using Census of India 2011 data in R.

Urbanisation is strongly right-skewed. Reading the bars, the lowest holds about three districts, the next 14 (roughly 15 to 25 per cent), then eight (25 to 35) and five (35 to 45). Then there is a gap, with one district in each of the 55–65, 65–75 and 75–85 bands and two at about 100 per cent. Household size, by contrast, is spread fairly evenly between 4.0 and 5.2 persons, so a log transformation adds little. In panel (b) the fitted line slopes gently downward, but the scatter is wide ($r = -0.14$, not significant). The two districts at 100 per cent urban have household sizes of about 4.4 and 4.6, close to the state middle, so metropolitan districts do not have particularly small households here.

Additional household-size plots. The accompanying R outputs examine household size on its original scale and after a logarithmic transformation, and compare both versions with urbanisation. The original household-size values span approximately 4.0–5.2, while the logged values span approximately 1.35–1.65. The logarithm compresses the scale, but the histogram alone does not show that the transformation improves the analysis. The scatterplots show substantial variation at similar urbanisation levels, so any apparent relationship should be interpreted cautiously.

Figure 12A. Household size and log-transformed household size: scatterplots against urbanisation and histograms. Source: author's analysis in R.



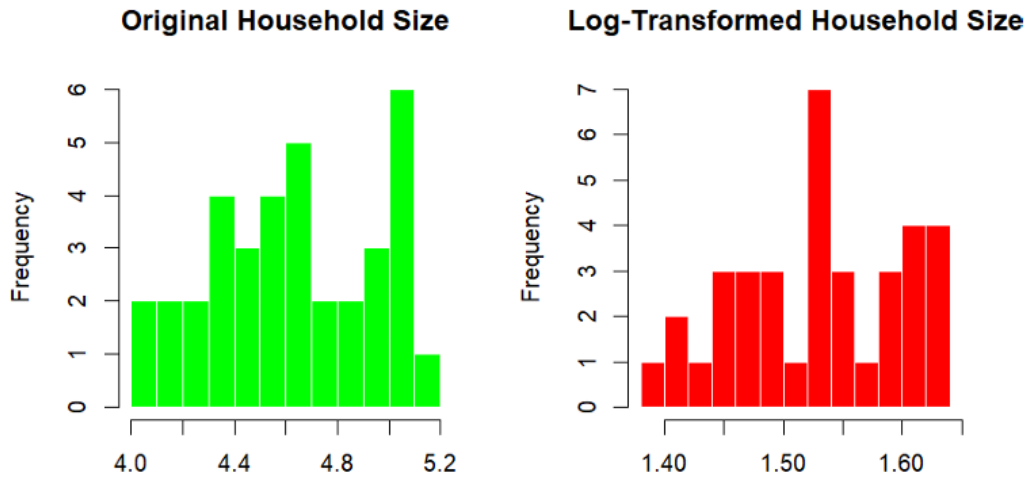
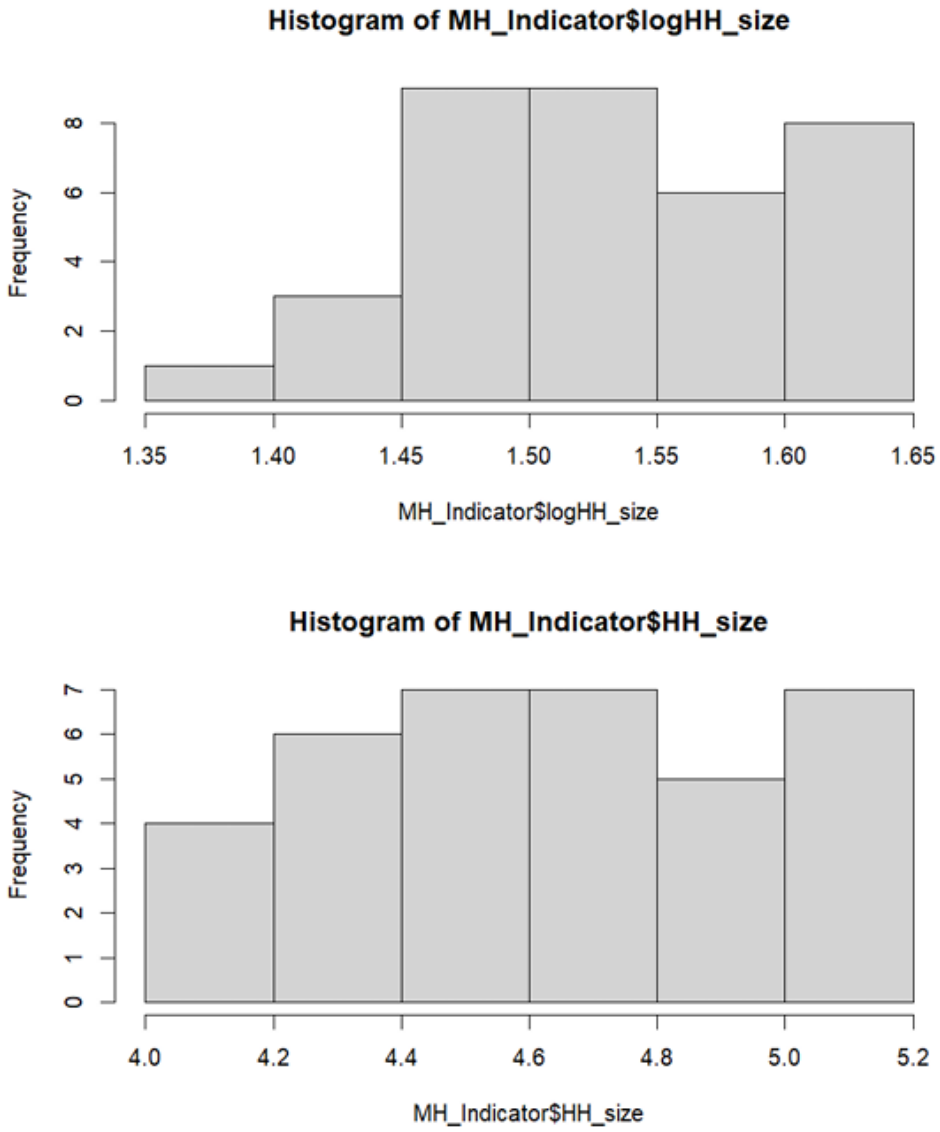
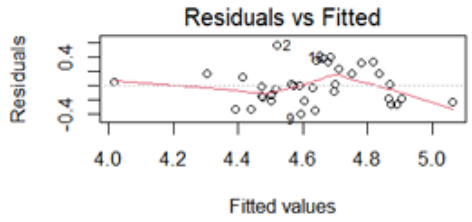


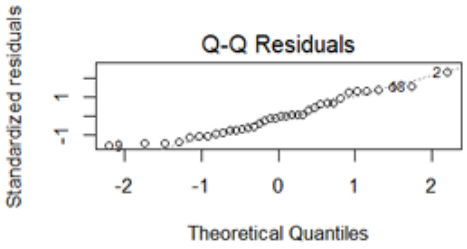
Figure 12B. Histograms of household size and log-transformed household size. Source: author's analysis in R.

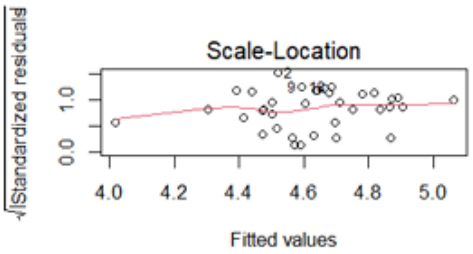


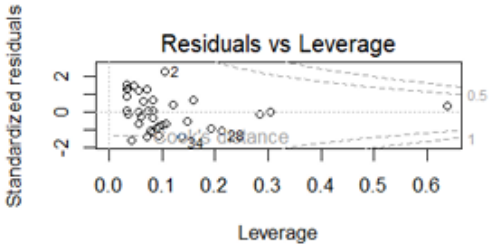
Exploratory regression diagnostics. The additional diagnostic panel below plots residuals against fitted values, a normal Q–Q plot, scale-location, and residuals against leverage. The fitted-value axis is around 4.0–5.1, which is consistent with a household-size outcome rather than the total work participation model estimated in Section 11. These plots are therefore treated as diagnostics for the exploratory household-size analysis and are not used as evidence about the OLS model for total WPR. The report's formal OLS diagnostics for total WPR remain those reported in Section 11.1.

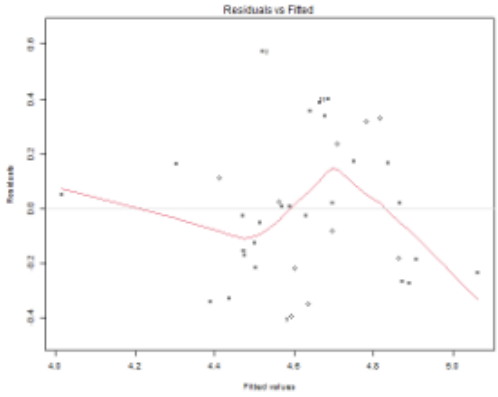
Figure 12C. Exploratory regression diagnostic plots for household size. Source: author's analysis in R.

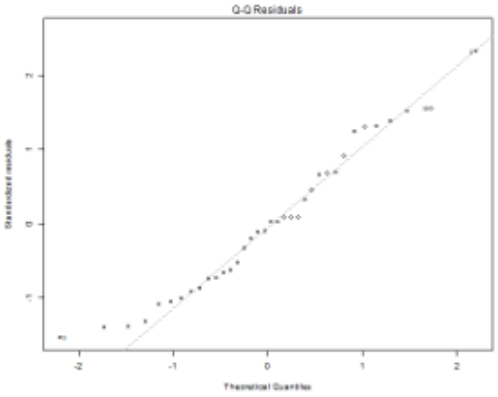

Residuals vs Fitted
Residuals
Fitted values

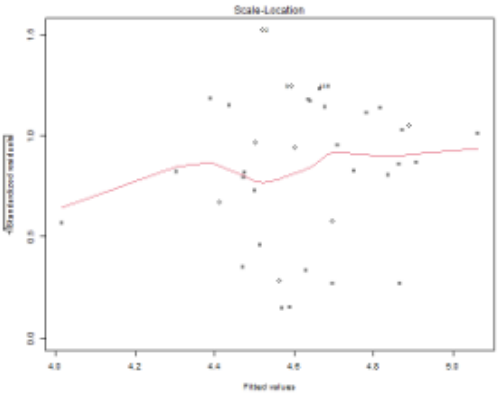

Q-Q Residuals
Standardized residuals
Theoretical Quantiles

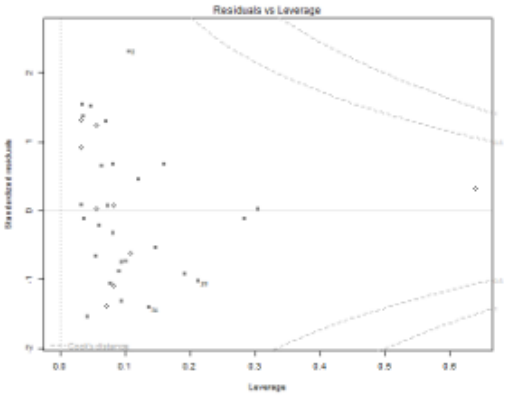

Scale-Location
√|Standardized residuals|
Fitted values


Residuals vs Leverage
Standardized residuals
Cook's distance
Leverage


Residuals vs Fitted


Q-Q Residuals


Scale-Location


Residuals vs Leverage

7. Step 4: Correlation Analysis

Pearson correlations were computed among the district indicators available in R. Figure 13 shows the correlation matrix. The discussion focuses on the demographic and work-participation relationships relevant to the research objectives. Significance symbols follow the plotting function's convention; the original R script was not retained to independently verify the exact star thresholds, so the coefficients and reported p-values should be read alongside the output rather than inferred from stars alone. With 35 districts, the sample is small and correlation estimates may be sensitive to individual districts.

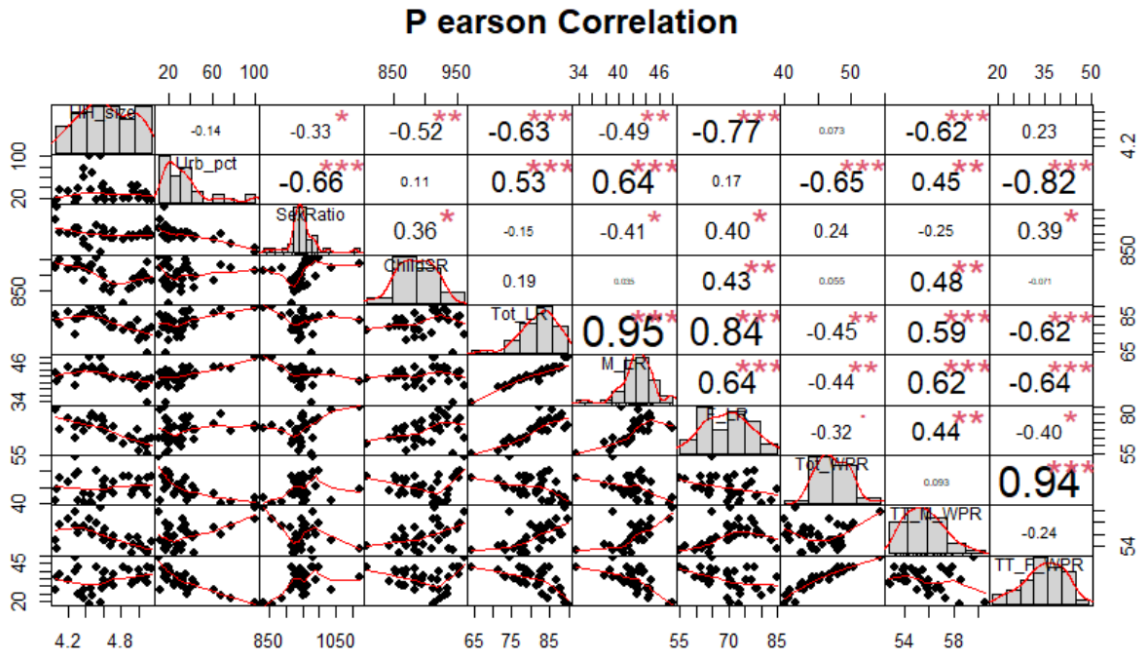


Figure 13. Pearson correlation matrix of ten district indicators, Maharashtra, Census 2011 (n = 35). Source: author's analysis in R. The literacy rows and columns are shown for completeness and are not discussed.

Pair	r	Significance
Total WPR – female WPR	+0.94	$p < 0.001$
Urbanisation – female WPR	−0.82	$p < 0.001$
Urbanisation – sex ratio	−0.66	$p < 0.001$
Urbanisation – total WPR	−0.65	$p < 0.001$

Pair	r	Significance
Household size – male WPR	−0.62	$p < 0.001$
Household size – child sex ratio	−0.52	$p < 0.01$
Child sex ratio – male WPR	+0.48	$p < 0.01$
Urbanisation – male WPR	+0.45	$p < 0.01$
Sex ratio – female WPR	+0.39	$p < 0.05$
Sex ratio – child sex ratio	+0.36	$p < 0.05$
Household size – sex ratio	−0.33	$p < 0.05$
Male WPR – female WPR	−0.24	not significant
Household size – female WPR	+0.23	not significant
Child sex ratio – urbanisation	+0.11	not significant
Total WPR – male WPR	+0.09	not significant
Household size – total WPR	+0.07	not significant

Table 2. Selected Pearson correlations (n = 35), taken from Figure 13.

Three results stand out.

1. **Total work participation is a story about women.** Total WPR is almost perfectly correlated with female WPR (0.94) and uncorrelated with male WPR (0.09). This agrees with the ranges in Section 5.3 and means any model of total WPR is largely a model of female participation.
2. **More urban districts have lower female and higher male participation.** Urbanisation is strongly negative with female WPR (−0.82) and moderately positive with male WPR (+0.45). The result is that urbanisation and total WPR are negatively correlated (−0.65). I do not read this as urbanisation reducing women's work. It may reflect the kinds of work that urban areas record, and the Mumbai districts at 100 per cent urban could carry much of the correlation. I have not tested that.
3. **Sex ratio and child sex ratio are related but different.** The correlation is only 0.36, and the child sex ratio is uncorrelated with urbanisation (0.11) while the sex ratio is strongly negative with it (−0.66). A district can therefore have a low overall sex ratio, for instance because of male in-migration, and still have a normal child sex ratio.

Household size is negatively correlated with male WPR (−0.62). This fits the maps. Marathwada districts such as Nanded, Parbhani and Latur have both large households and the lowest male

participation. Correlations at district level describe places and not individuals, so they do not show that large households cause low participation.

8. Step 5: Defining Neighbours

Every spatial statistic depends on the definition of a neighbour. In GeoDa, the weights file named “GIS project Pravesh” was used for the 35 districts. The project notes identify the weights as Queen contiguity, first order, with a symmetric specification. Figure 14 shows the connectivity graph. The graph should be read as a diagnostic of spatial connections, not as proof that every polygon boundary is geometrically correct. The small Mumbai polygons warrant particular caution when interpreting local clusters.

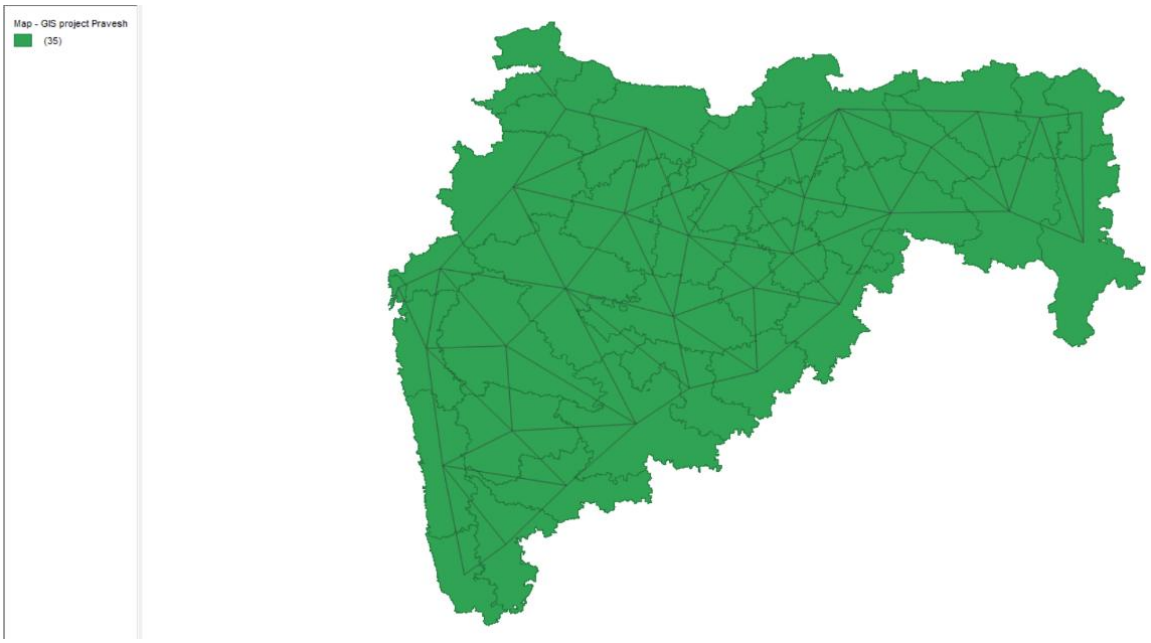


Figure 14. Connectivity graph of the district spatial weights (35 districts), GeoDa. Source: author's analysis.

The connectivity graph is used to inspect the spatial links for the 35-district layer. The Mumbai polygons are very small at the state scale, so their exact geometry and links are difficult to assess visually. This is a limitation of the available map evidence and is considered when discussing results involving Mumbai and neighbouring coastal districts.

9 Step 6: Global Spatial Autocorrelation (Moran's I)

Table 3 gives Global Moran's I for the seven non-literacy indicators in the GeoDa output. All used 999 permutations, so the smallest possible pseudo p-value is 0.001, and that is what every test returned. The expected value under randomness is −0.0294 throughout.

Indicator	Moran's I	z-value	Pseudo p
Household size	0.606	5.26	0.001
Male WPR	0.599	5.32	0.001
Urbanisation	0.564	5.42	0.001
Child sex ratio	0.557	5.03	0.001
Sex ratio	0.533	5.35	0.001
Female WPR	0.508	4.71	0.001
Total WPR	0.431	4.21	0.001

Table 3. Global Moran's I, spatial weights as in Section 8, 999 permutations, n = 35. Source: author's analysis in GeoDa.

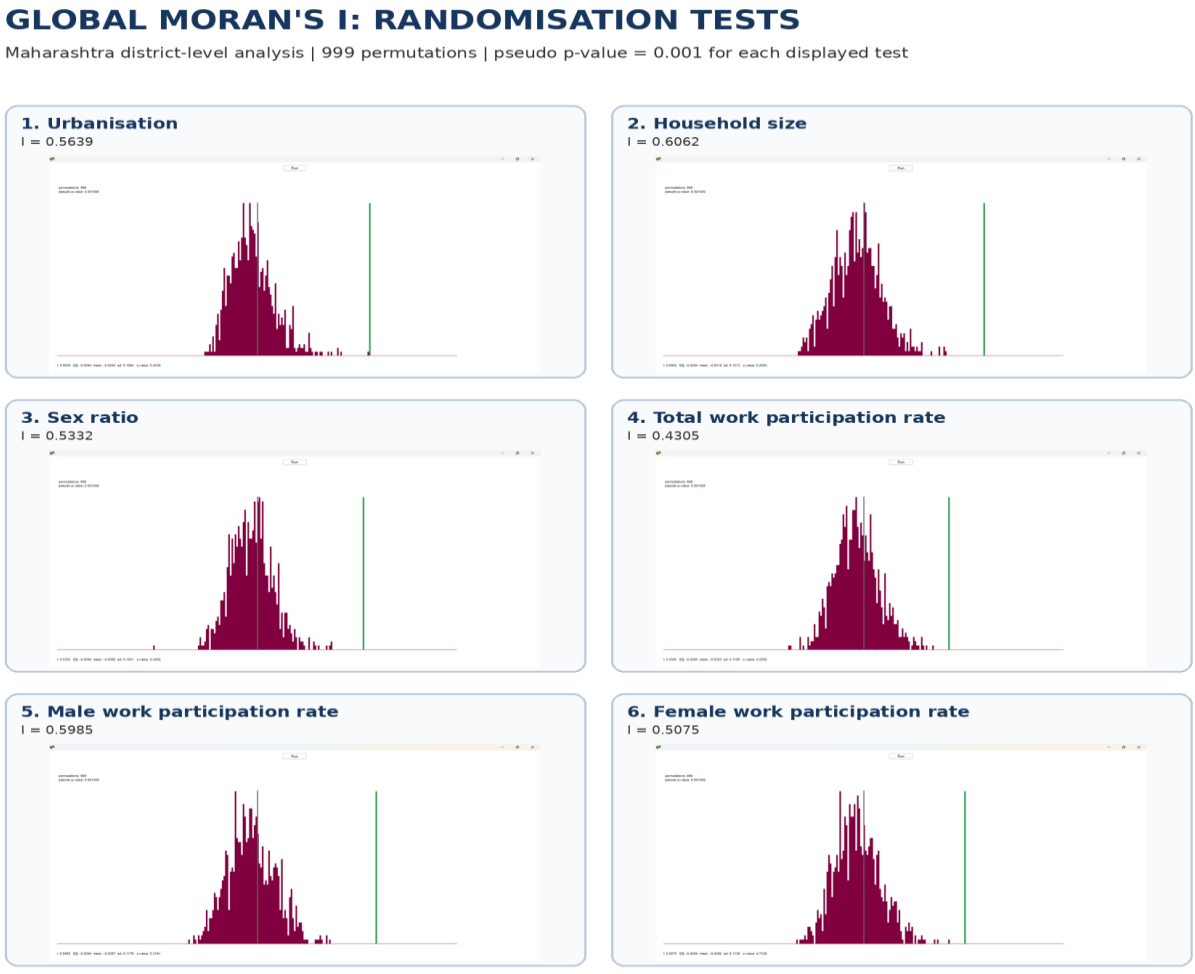


Figure 15. Randomisation tests for Global Moran's I across six demographic and work-related indicators in Maharashtra. Source: Author's analysis in GeoDa using 999 permutations.

The tests report pseudo p-values of 0.001 for all six indicators, indicating statistically significant positive spatial autocorrelation. Household size has the highest observed Moran's I (0.6062), followed by male work participation (0.5985) and urbanisation (0.5639). Total work participation has the lowest value among the six indicators (0.4305), but remains statistically significant. Overall, the results suggest that districts with similar values tend to be located near one another. The LISA analysis identifies the locations of specific local clusters and spatial outliers.

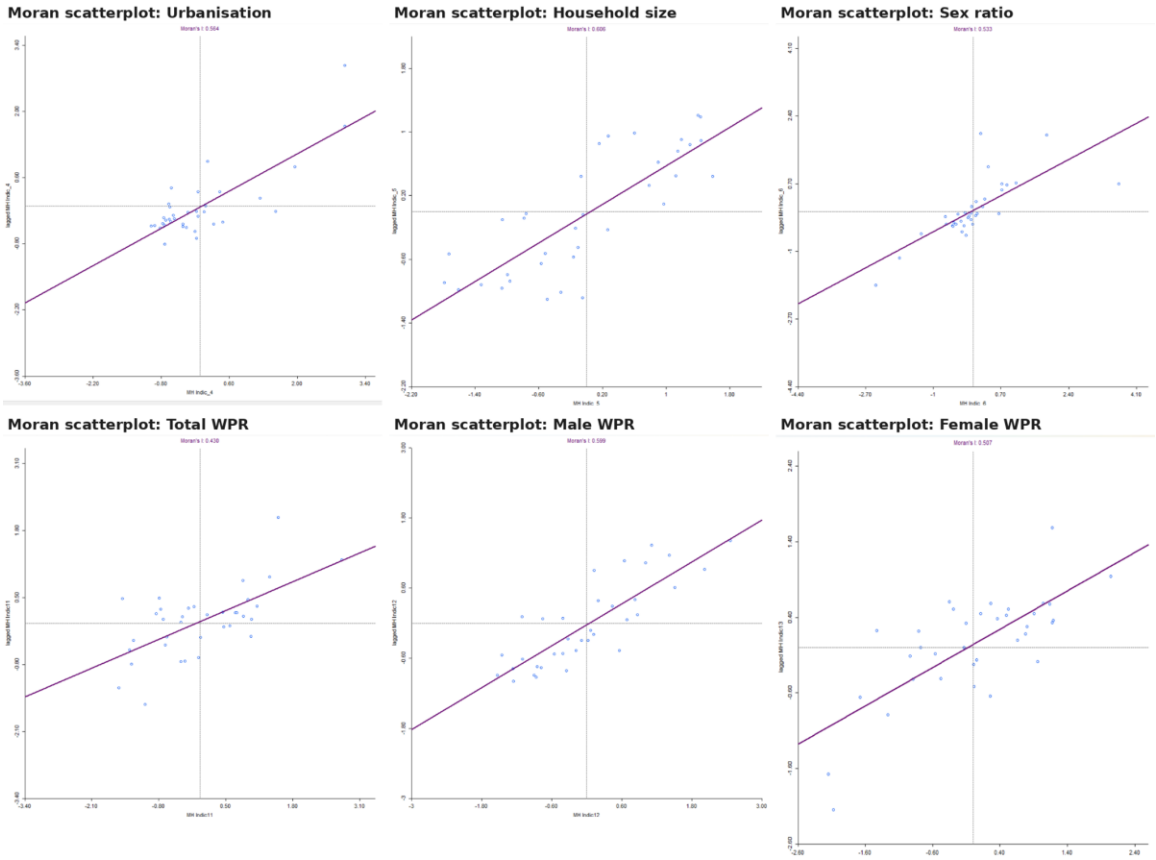


Figure 16. Moran scatterplots for six indicators (standardised value on the horizontal axis, spatial lag on the vertical axis). The slope of the line is Moran's I. Source: author's analysis in GeoDa.

Every indicator is positively and significantly autocorrelated. Moran's I runs from 0.43 to 0.61, against an expected value near zero, and the observed value sits well outside the permutation distribution in each case (z between 4.2 and 5.4). Household size is the most clustered indicator (0.606), followed closely by male WPR (0.599) and urbanisation (0.564). Total WPR is the least clustered (0.431), though still strongly so. In every scatterplot the fitted line slopes upward and most districts are in the High-High or Low-Low quadrants.

Reading Figure 15. Each panel plots a district's standardised value on the horizontal axis against the average standardised value of its neighbours on the vertical axis. The upper-right quadrant holds High-High districts (above the mean, with above-average neighbours) and the lower-left quadrant holds Low-Low districts. The upper-left and lower-right quadrants hold the possible outliers, Low-High and High-Low. The slope of the line is Moran's I, so a steep line with most points in the two diagonal quadrants means a strongly clustered indicator. Not every district in a diagonal quadrant is a significant cluster. LISA colours only those that pass the permutation test at $p \leq 0.05$, which is why the maps in Section 10 show between six and sixteen coloured districts for each indicator and not every district in those quadrants.

Two cautions apply. First, a large positive Moran's I says that similar districts are near each other, not why. Second, with 35 districts and a few extreme observations, such as the Mumbai districts for urbanisation, a handful of places can raise the statistic. Section 10 shows which districts are behind it.

10. Step 7: Local Clusters (LISA)

Global Moran's I says the map is clustered. LISA says where. I mapped districts with $p \leq 0.05$ (999 permutations) for six indicators. Table 4 counts the clusters and Figures 16 to 21 show them. The GeoDa maps carry no district labels. I named the districts by matching their outlines to the labelled QGIS maps, and in every map the number of High-High, Low-Low and Low-High districts I found equals the count in the GeoDa legend. Where two small adjacent polygons cannot be told apart (Mumbai City and Mumbai Suburban) I say so. The names should still be confirmed against GeoDa's saved cluster table.

Indicator	High-High	Low-Low	Low-High	High-Low	Not significant	Where the clusters are
Urbanisation	4	2	0	0	29	High-High: Thane, Mumbai Suburban, Mumbai City, Raigad. Low-Low: Gondia, Kolhapur.
Household size	9	7	0	0	19	High-High: Dhule, Jalgaon, Aurangabad, Jalna, Parbhani,

Indicator	High-High	Low-Low	Low-High	High-Low	Not significant	Where the clusters are
						Bid, Nanded, Latur, Osmanabad. Low-Low: Raigad, Kolhapur, Wardha, Nagpur, Bhandara, Chandrapur, Gadchiroli.
Sex ratio	4	3	0	0	28	High-High: Ratnagiri, Sindhudurg, Kolhapur, Sangli. Low-Low: Ahmednagar, Mumbai Suburban, Mumbai City.
Total WPR	4	4	0	0	27	High-High: Bhandara, Gondia, Gadchiroli, Chandrapur. Low-Low: Mumbai Suburban, Mumbai City, Raigad, Ratnagiri.

Indicator	High-High	Low-Low	Low-High	High-Low	Not significant	Where the clusters are
Male WPR	6	7	0	0	22	High-High: Nagpur, Bhandara, Gondia, Chandrapur, Gadchiroli and one Mumbai district. Low-Low: Jalgaon, Aurangabad, Jalna, Parbhani, Bid, Latur, Osmanabad.
Female WPR	3	4	1	0	27	High-High: Gondia, Gadchiroli, Hingoli. Low-Low: Thane, Mumbai Suburban, Mumbai City, Raigad. Low-High: Aurangabad.

Table 4. LISA cluster counts at $p \leq 0.05$ (999 permutations, $n = 35$) and where the clusters lie. Source: author's analysis in GeoDa.

10.1 How to read a cluster map

A red district is High-High: its value is above the state mean and so is the average of its neighbours. A blue district is Low-Low, a light blue one is Low-High (a low value among high neighbours) and a pink one would be High-Low. Grey districts are not significant at $p \leq 0.05$

with 999 permutations. A cluster is therefore a statement about a district and its neighbours together. A district can be in the top class on the QGIS map and not be flagged because its neighbours differ, and a district in a middle class can be flagged because everything around it is high or low. Each subsection below gives the districts in each cluster, how they compare with the QGIS class of the same indicator, and what can and cannot be said.

10.2 Urbanisation (Figure 16)

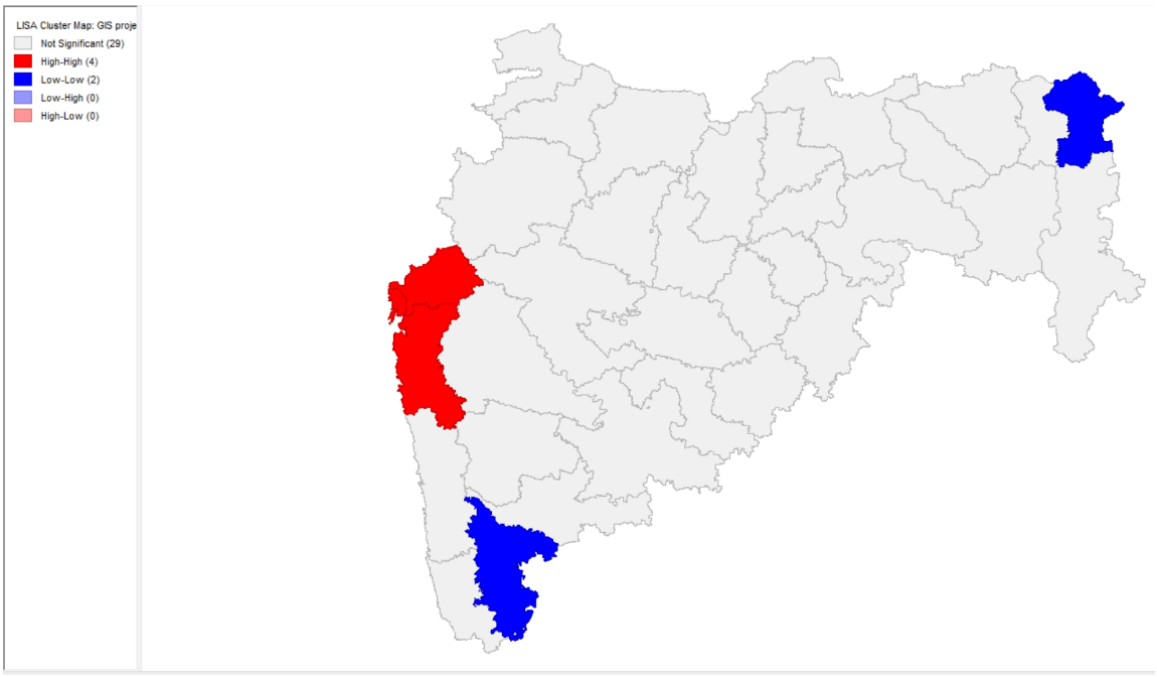


Figure 16. LISA cluster map for urbanisation (red: High-High, blue: Low-Low; $p \leq 0.05$, 999 permutations). Source: author's analysis in GeoDa.

Result. Four High-High and two Low-Low districts, with 29 not significant. High-High: Thane, Mumbai Suburban, Mumbai City and Raigad, one connected block on the northern Konkan coast. Low-Low: Gondia in the far north-east and Kolhapur in the south-west, two isolated single-district clusters.

Reading. The High-High block is the Mumbai metropolitan region. Pune and Nagpur are also in the 44–77 per cent class on the QGIS map (Figure 1), but neither is a cluster, because their neighbours are mostly in lower urbanisation classes. They are urban islands and not part of a spatial block. This sharpens the global result: Moran's I of 0.56 summarises the whole map, and the significant local pattern is the Mumbai coast. Gondia is in the lowest class and so are its neighbours Bhandara and Gadchiroli, although those two are not flagged. Kolhapur is in the 22–33 class and is flagged because Sindhudurg, Ratnagiri and Satara next to it are in the lowest class.

Caution. The two Mumbai polygons are tiny, and the High-High result depends on how they are linked to Thane and Raigad (Section 8).

10.3 Household size (Figure 17)

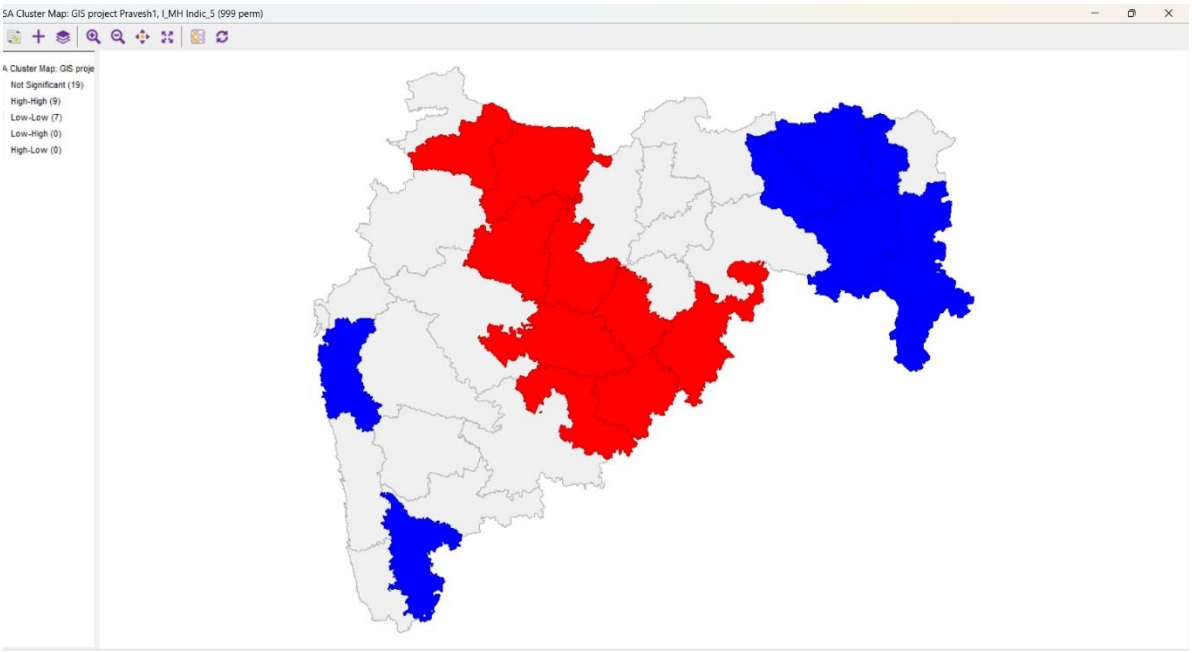


Figure 17. LISA cluster map for household size. Source: author's analysis in GeoDa.

Result. Nine High-High and seven Low-Low districts, the most clustered map in the report (19 not significant). High-High: Dhule, Jalgaon, Aurangabad, Jalna, Parbhani, Bid, Nanded, Latur and Osmanabad. Low-Low: Raigad, Kolhapur, Wardha, Nagpur, Bhandara, Chandrapur and Gadchiroli.

Reading. The High-High districts form one connected block from the north-west across Marathwada. It is not the same as the top QGIS class. Jalgaon and Osmanabad are in the middle class (4.47–4.72) and are flagged because their neighbours have large households. Nandurbar, Nashik, Hingoli, Ahmednagar and Solapur are in the top two classes but are not flagged. The Low-Low districts fall in two groups: five contiguous districts in eastern Vidarbha (Wardha, Nagpur, Bhandara, Chandrapur and Gadchiroli) and two in the south-west and on the coast (Kolhapur and Raigad). The map agrees with the Moran's I of 0.61, the highest of the ten indicators. The contrast between the Marathwada and eastern Vidarbha blocks is the sharpest in the dataset. The map shows where households are large or small, not why.

10.4 Sex ratio (Figure 18)

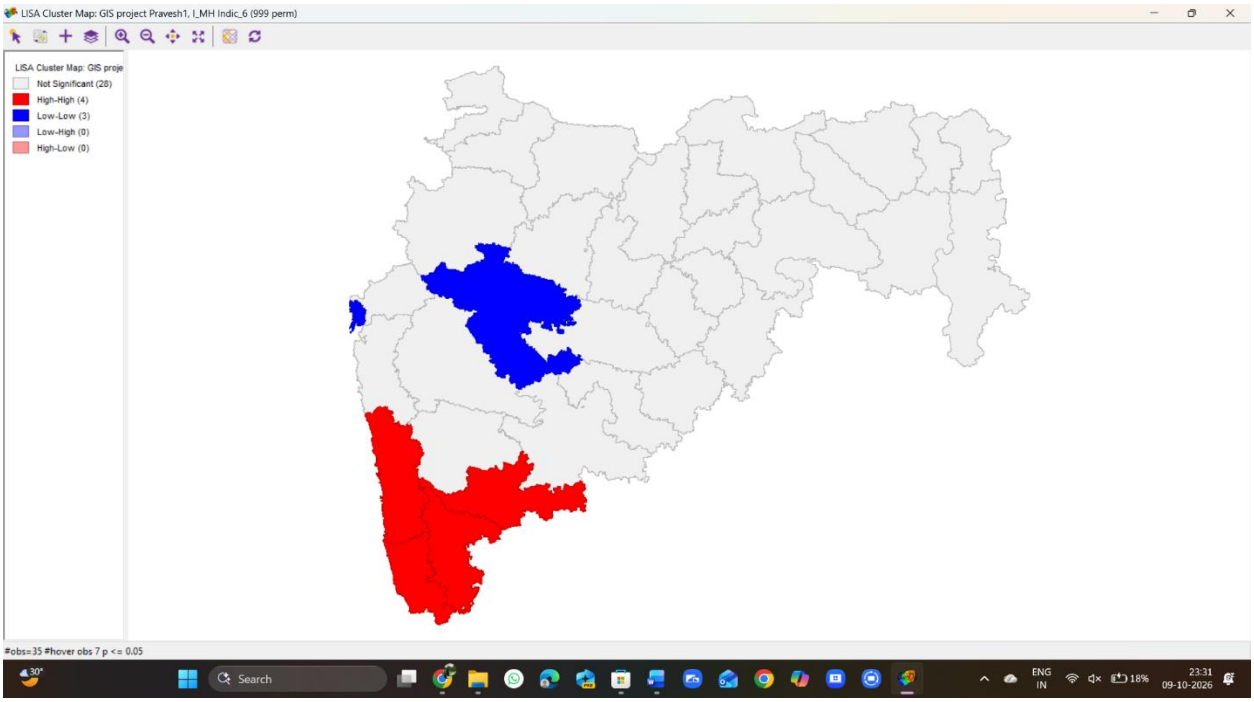


Figure 18. LISA cluster map for sex ratio. Source: author's analysis in GeoDa.

Result. Four High-High and three Low-Low districts. High-High: Ratnagiri, Sindhudurg, Kolhapur and Sangli, a connected block in the south-west. Low-Low: Ahmednagar and the two Mumbai districts.

Reading. Ratnagiri has the highest sex ratio in the state and Sindhudurg is in the second-highest class. Kolhapur and Sangli are in the middle class (937–966) and are flagged because they border these districts. Satara is in the second-highest class but is not flagged. The Mumbai pair is in the lowest QGIS class, as is Thane, which is not flagged. Ahmednagar is in the middle class itself and is flagged because most of its neighbours are in a lower one. The map sets a south-western high against a Mumbai and north-central low. The data show where this happens. They do not show whether the cause is demographic, migration-related or something else.

10.5 Total work participation (Figure 19)

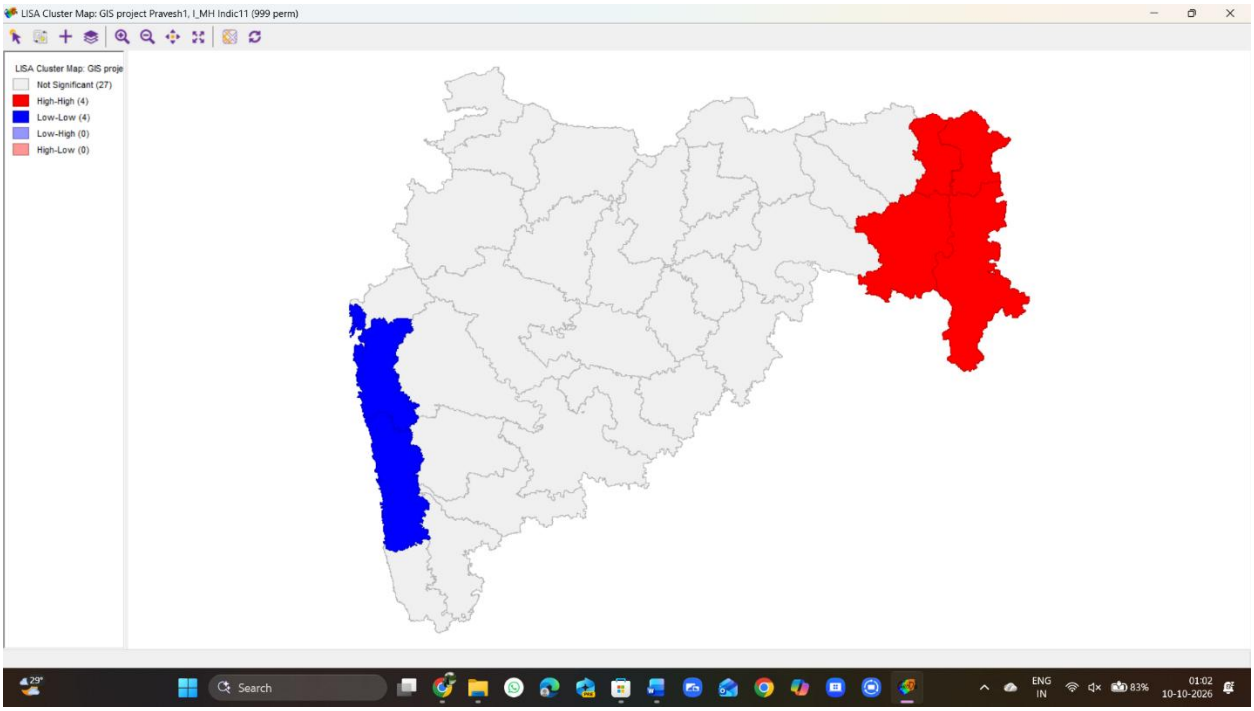


Figure 19. LISA cluster map for total work participation rate. Source: author's analysis in GeoDa.

Result. Four High-High and four Low-Low districts. High-High: Bhandara, Gondia, Gadchiroli and Chandrapur, the north-eastern corner of Vidarbha. Low-Low: Mumbai Suburban, Mumbai City, Raigad and Ratnagiri, the Konkan coast.

Reading. This is the cleanest east–west contrast in the report: two clusters at opposite ends of the state. Nagpur is in the lowest class on the QGIS map but is not a cluster, because its neighbours Bhandara, Chandrapur and Wardha are higher. Thane and Sindhudurg are also in the lowest class and are not flagged. Ratnagiri is flagged although it is in the middle class, because the coast around it is low. Total WPR has the lowest Moran's I of the ten indicators (0.43), and the cluster map shows why: only two compact regions are significant and the rest of the state is mixed.

10.6 Male work participation (Figure 20)

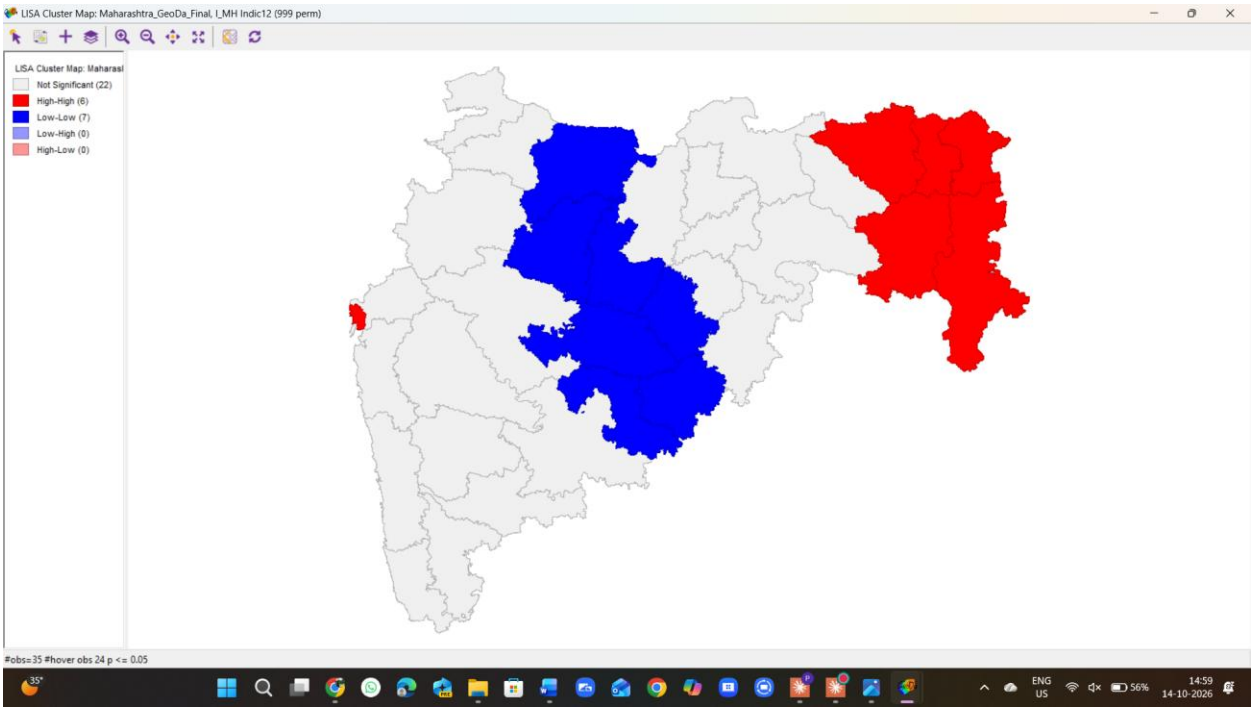


Figure 20. LISA cluster map for male work participation rate. Source: author's analysis in GeoDa.

Result. Six High-High and seven Low-Low districts. High-High: Nagpur, Bhandara, Gondia, Chandrapur and Gadchiroli in eastern Vidarbha, and one small polygon at Mumbai. Low-Low: Jalgaon, Aurangabad, Jalna, Parbhani, Bid, Latur and Osmanabad.

Reading. The Low-Low districts form a continuous belt through Marathwada and Jalgaon. All seven are also High-High for household size (Section 10.3), which is the spatial counterpart of the correlation of −0.62 between household size and male WPR. Osmanabad is in the middle QGIS class and Jalgaon in the second-lowest, so both are flagged through their neighbours. Nanded, in the lowest class, is not flagged. In the east, Nagpur and Bhandara are in the middle class on the QGIS map and are flagged because the districts around them are high, while Wardha, in the top class, is not. Four of the five eastern districts (Nagpur, Bhandara, Chandrapur and Gadchiroli) are Low-Low for household size, so eastern Vidarbha combines small households and high male participation.

Caution. The Mumbai cluster is a single tiny polygon. The screenshot does not show whether it is Mumbai City or Mumbai Suburban (Mumbai Suburban is in the top class in Figure 5), and I would not rely on it.

10.7 Female work participation (Figure 21)

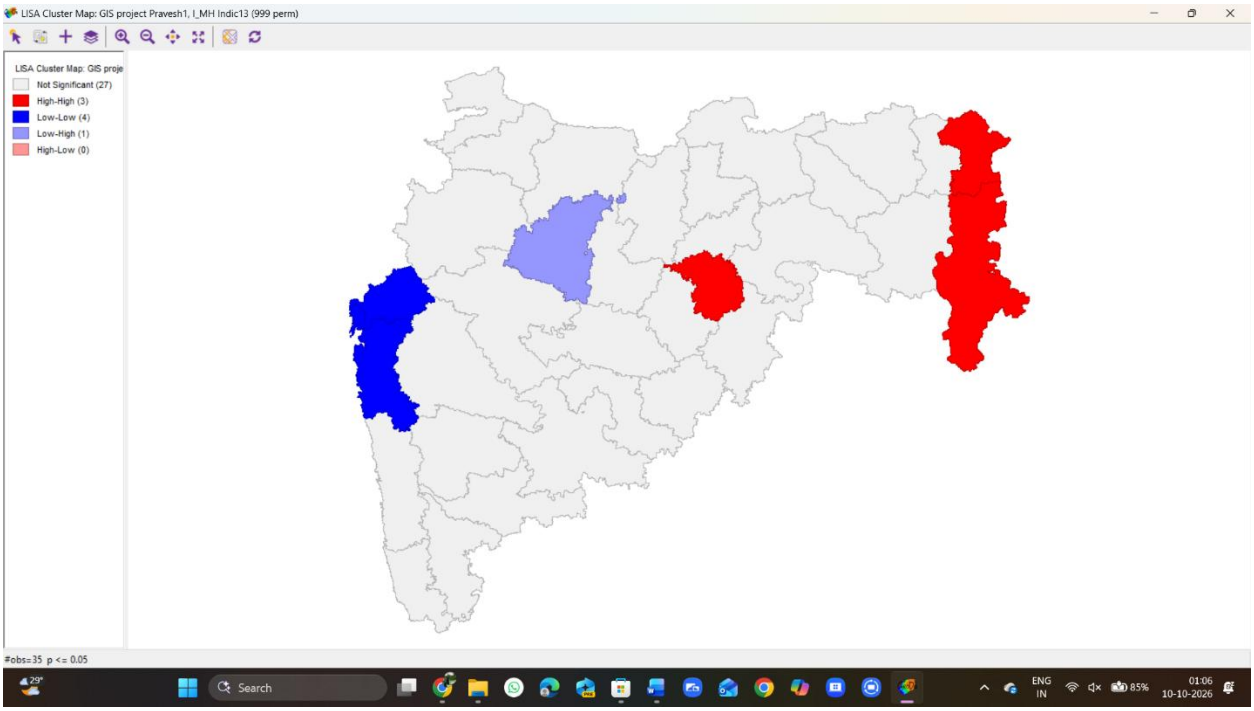


Figure 21. LISA cluster map for female work participation rate (light blue: Low-High). Source: author's analysis in GeoDa.

Result. Three High-High, four Low-Low and one Low-High district. High-High: Gondia, Gadchiroli and Hingoli. Low-Low: Thane, Mumbai Suburban, Mumbai City and Raigad. Low-High: Aurangabad.

Reading. The Low-Low coast is the same four districts that are High-High for urbanisation (Figure 16). This is the most robust local finding in the report: one contiguous block is both the urban cluster and the low female participation cluster, and the correlation across all districts is −0.82. Gondia and Gadchiroli continue the north-eastern high-participation region of the total WPR map. Hingoli is an isolated High-High district in the second-highest QGIS class, and it is the only High-High that is not part of the eastern block. Aurangabad is the only Low-High district: its female WPR is in the second-lowest class while its neighbours Jalna, Ahmednagar, Bid and Buldana are in the second-highest. One outlier does not make a pattern and I do not interpret it further.

10.8 The six maps together

Three regions recur. First, the northern Konkan coast (Thane, Mumbai Suburban, Mumbai City and Raigad). The same four districts are High-High for urbanisation and Low-Low for female WPR. Raigad and the two Mumbai districts are also Low-Low for total WPR, the Mumbai pair is Low-Low for sex ratio, and Raigad is Low-Low for household size. Second, eastern Vidarbha (Gadchiroli, Gondia, Bhandara and Chandrapur). These are High-High for total and male WPR, Gadchiroli and Gondia are High-High for female WPR, and Gadchiroli, Bhandara and Chandrapur are Low-Low for household size. Gondia is also Low-Low for urbanisation. Third,

Marathwada: Aurangabad, Jalna, Parbhani, Bid, Latur and Osmanabad are High-High for household size and Low-Low for male WPR, and all seven male WPR Low-Low districts, Jalgaon included, are also in the household-size High-High cluster.

What differs. Sex ratio and participation do not line up. The south-western sex ratio cluster (Ratnagiri, Sindhudurg, Kolhapur and Sangli) shows up in no participation map, except that Ratnagiri is Low-Low for total WPR. Pune and Nagpur, the inland districts in the 44–77 per cent urban class, belong to no urbanisation cluster. Nagpur is in the lowest QGIS class for total and female WPR, and its only clusters are household size (Low-Low) and male WPR (High-High). There are no High-Low districts on any map.

Consistency with the correlations. The overlaps reproduce in space what Section 7 found across districts. The urbanisation and female WPR clusters coincide on the Konkan coast ($r = -0.82$), and the household size and male WPR clusters coincide in Marathwada ($r = -0.62$). In eastern Vidarbha the total and male WPR clusters coincide, even though the correlation between total and male WPR is only 0.09, because that correlation is dominated by the large differences in female participation elsewhere.

Caution. The number of districts is small, and testing 35 local statistics at a 5 per cent threshold without a multiple-testing adjustment can produce chance findings. The results also depend on the neighbour definition, and the small Mumbai polygons are difficult to inspect at state scale. No LISA map was included for child sex ratio, so no local-cluster interpretation is made for that indicator. Clusters show where similar values are located; they do not explain why.

11. Step 8: Modelling Total Work Participation

The correlations in Section 7 suggest that total WPR falls with urbanisation. I tested this with a regression of total WPR (MH Indic11) on urbanisation (MH Indic_4), average household size (MH Indic_5) and total literacy (MH Indic_8), which is included as a control only. There are 35 districts. I first fitted OLS, which ignores space, and then a spatial error model, which allows the unexplained part of the work rate to be correlated between neighbours. Both were estimated in GeoDa with the weights of Section 8.

Term	OLS coefficient (SE)	OLS p	Spatial error coefficient (SE)	Spatial error p
Constant	70.94 (16.57)	0.0002	65.59 (12.98)	< 0.001
Total literacy (Indic_8)	−0.176 (0.124)	0.167	−0.154 (0.101)	0.129
Urbanisation (Indic_4)	−0.0770 (0.0246)	0.004	−0.0945 (0.0243)	0.0001
Household size (Indic_5)	−1.926 (1.844)	0.304	−1.045 (1.651)	0.527

Term	OLS coefficient (SE)	OLS p	Spatial error coefficient (SE)	Spatial error p
Lambda (spatial error)	–	–	0.653 (0.133)	< 0.001
R-squared	0.451 (adj. 0.398)	–	0.659 (pseudo, not comparable)	–
Log-likelihood	−81.07	–	−75.12	–
AIC	170.14	–	158.24	–
Schwarz criterion	176.37	–	164.46	–
Sigma-squared	6.795	–	3.738	–

Table 6. OLS and spatial error estimates for total work participation (dependent variable MH Indic11, n = 35). Spatial lag estimates are presented in Table 6A. Source: author's analysis in GeoDa.

11.1 The OLS model and its diagnostics

The OLS model is significant overall ($F = 8.48$ on 3 and 31 degrees of freedom, $p = 0.0003$) and explains 45 per cent of the variation in total WPR (adjusted R^2 0.40). Only urbanisation has a significant coefficient: each additional percentage point of urban population goes with a 0.077 point lower total WPR, or about 0.8 points for a 10 point difference. Literacy and household size are not significant. They are correlated with each other (Section 7), and the multicollinearity condition number of 90 is high, so their separate effects are poorly pinned down.

The residuals are close to normal (Jarque-Bera 0.045, $p = 0.978$), but all three heteroskedasticity tests reject constant variance at the 5 per cent level (Breusch-Pagan 10.28, $p = 0.016$; Koenker-Bassett 9.77, $p = 0.021$; White 20.28, $p = 0.016$). OLS standard errors are therefore unreliable. Heteroskedasticity can be a sign of a misspecified or spatially structured model, which is what the next model addresses.

11.2 The spatial error model

The spatial error coefficient is positive and statistically significant (lambda = 0.653, $z = 4.91$, $p < 0.001$). The likelihood-ratio test of the spatial error model against OLS is significant ($p = 0.0006$). The AIC falls from 170.14 for OLS to 158.24 for the spatial error model, and the standard error of regression falls from 2.61 to 1.93. The Breusch–Pagan test for the spatial error model is not significant ($p = 0.391$). Urbanisation remains the only statistically significant explanatory variable; total literacy and household size are not significant.

11.3 The Spatial Lag Model

The spatial lag model includes the spatially lagged dependent variable to account for dependence in total work participation across neighbouring districts. The estimated spatial autoregressive coefficient is positive (rho = 0.426, p = 0.010), and the likelihood-ratio test is significant (p = 0.0148). The model AIC is 166.20, lower than the OLS AIC of 170.14 but higher than the spatial error AIC of 158.24. Urbanisation remains negative and statistically significant (coefficient = −0.055, p = 0.015); total literacy and household size are not statistically significant. The spatial lag coefficient represents spatial feedback and should not be read as a direct marginal effect of the explanatory variables.

Table 6A. Spatial lag model for total work participation (dependent variable MH Indic11, n = 35). Source: author's analysis in GeoDa.

Term	Coefficient	p-value
Constant	50.218	0.0018
Total literacy (MH Indic_8)	−0.1678	0.1093
Urbanisation (MH Indic_4)	−0.0553	0.0152
Household size (MH Indic_5)	−1.9209	0.2165
Spatial lag (rho)	0.4263	0.0102
AIC	166.198	—
Sigma-squared	4.828	—
Likelihood-ratio test for spatial lag	—	0.0148

Model comparison. The OLS, spatial lag and spatial error models have AIC values of 170.14, 166.20 and 158.24, respectively. The spatial error model also has the smallest reported standard error of regression (1.93, compared with 2.61 for OLS and 2.20 for the spatial lag model). Taken together, these criteria favour the spatial error model among the three specifications estimated. However, the saved outputs did not include the full Lagrange multiplier diagnostics or Moran's I for OLS residuals, so this should be treated as a reasoned comparison rather than a complete diagnostic selection process.

Figure 22. Histograms and normal Q-Q plots of OLS residuals (top) and spatial error residuals (bottom). Source: author's analysis in R. Some plot titles are cut off in the source output.

The residual plots show the narrower spread under the spatial error model, with OLS residuals ranging from about −8 to +5 and spatial error residuals from about −5 to +4. Both are roughly normal.

11.4 What This Does and Does Not Show

The result says that, after allowing for urbanisation, household size and literacy, the unexplained part of total WPR is itself clustered in space. Neighbouring districts share influences that the model omits. I can name plausible candidates, such as agro-ecology, the structure of local labour markets and tribal composition, but I have not tested any of them. The negative urbanisation coefficient is an association between districts and not evidence that urbanisation lowers work participation.

Diagnostic limitation. The saved outputs did not include the Lagrange multiplier tests or Moran's I of the OLS residuals. Therefore, model selection is based on the reported coefficient estimates, likelihood-ratio tests and fit statistics. The spatial error model is preferred by AIC and residual standard error among the three estimated specifications, but the comparison is not a complete diagnostic-selection procedure.

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12. Discussion

The evidence supports three levels of claim, and I keep them apart.

What the data directly show

- Urbanisation, household size, sex ratio, child sex ratio and the three work participation rates are all significantly clustered in space (Moran's I from 0.43 to 0.61, pseudo p = 0.001).
- Total work participation tracks women's participation (r = 0.94) and not men's (r = 0.09). The female rate varies about four times as widely across districts as the male rate.
- Urbanisation is negatively correlated with female WPR (−0.82), sex ratio (−0.66) and total WPR (−0.65), and positively with male WPR (+0.45).
- In the regression of total WPR, urbanisation is the only explanatory variable that remains statistically significant across OLS, spatial lag and spatial error specifications. The spatial error coefficient is strongly significant (lambda = 0.653, p < 0.001), and the spatial error model has the lowest AIC of the three models. This indicates that spatial dependence in omitted influences is important, although the analysis does not identify those influences or establish causation.
- LISA places high total work participation in Bhandara, Gondia, Gadchiroli and Chandrapur, low total participation in Mumbai Suburban, Mumbai City, Raigad and Ratnagiri, low female participation in Thane, Mumbai and Raigad (the same districts that form the urbanisation cluster), and a block of large households and low male participation across Aurangabad, Jalna, Parbhani, Bid, Latur and Osmanabad.

What can reasonably be interpreted

The maps and clusters sketch three regional profiles. The western metropolitan belt (Mumbai, Thane and Raigad, with Pune in a weaker form) is highly urban. Thane and Mumbai have the

lowest sex ratios, and Thane, Mumbai and Raigad the lowest female participation, while Pune is one class higher on both. Household sizes in Thane, Raigad and Pune are in the second-smallest class. Marathwada is mostly rural, with large households, high cultivator shares in Jalna, Bid and Hingoli, and the lowest male work participation in Aurangabad, Parbhani, Nanded and Latur. Eastern Vidarbha and the tribal margins, with Gadchiroli the clearest case, combine very low urbanisation, high ST shares, high sex ratios and the highest recorded participation by women. Nandurbar, the district with the highest ST share (up to 69 per cent), has large households, a high sex ratio and the lowest male participation class. These are descriptions of how indicators coincide in places. They do not tell us which one drives which.

Some patterns are consistent across indicators and some are not. Gadchiroli stands out on several maps and in several clusters. It is in the top class for total, male and female work participation, it is High-High for all three, and it is Low-Low for household size. Nagpur does not behave as the Mumbai–Thane belt does, however: it is in the lowest female-WPR class but not in a cluster, because its neighbours are not low. Sex ratio and work participation also pull apart on the south-western coast.

What remains a hypothesis

I cannot tell from these data why the patterns exist. Plausible explanations include male in-migration to the cities (for the low urban sex ratios), differences in how women's work is recorded in urban and rural areas, the agro-ecological and livelihood base of tribal and forested districts, and the structure of local labour markets. To test them I would need migration tables, the split of main and marginal workers, occupation by sex, and district-level economic data. None of these is in the material I have.

13. Conclusion

This report asked whether the demographic and work indicators of Maharashtra's 35 districts in 2011 were spatially random or organised into regional patterns. The seven indicators reported in the main Global Moran's I table show positive spatial autocorrelation, with Moran's I ranging from 0.431 for total WPR to 0.606 for household size; each has a pseudo p-value of 0.001 from 999 permutations. These results indicate spatial clustering under the weights used, not a causal process.

The main gaps are between regions and between genders. Women's work participation ranges from 18 to 49 per cent across districts, against 53 to 61 per cent for men, and total participation follows the female rate almost exactly. It is highest around Gadchiroli in the east and lowest along the northern Konkan coast, where the cluster analysis confirms Low-Low blocks: Mumbai Suburban, Mumbai City, Raigad and Ratnagiri for total participation, and Thane, Mumbai Suburban, Mumbai City and Raigad for female participation. Sex ratios run from 832 in the Mumbai–Thane area to 1,122 in Ratnagiri. Urbanisation is concentrated in a western belt and at Nagpur, with most districts below 45 per cent.

In the regression, urbanisation is the only explanatory variable that is statistically significant across the three specifications. Its coefficient is negative in OLS (−0.077), spatial lag (−0.055)

and spatial error (−0.095). The spatial error model has the lowest AIC (158.24), compared with 166.20 for the lag model and 170.14 for OLS. The spatial error parameter is positive and significant (lambda = 0.653, $p < 0.001$), suggesting spatial dependence in unobserved influences. These are district-level associations and should not be read as causal effects.

For planning, the findings suggest that district and state averages can hide regional blocks. The patterns in Marathwada, the Konkan coast and eastern Vidarbha support the value of region-sensitive analysis. Further work should preserve the original Census table references and boundary metadata, verify the operational definitions of all indicators, test sensitivity to alternative spatial weights, and incorporate additional labour-market, migration and socioeconomic variables.

14. Limitations

- **Old data and a changed map.** The Census is from 2011, and Maharashtra now has 36 districts. The results describe the 2011 state and cannot be taken as current conditions.
- District-level exact values. The report's district classifications and some descriptions were read from the QGIS maps and legends, while the full cleaned district attribute table was not retained in the report package. Consequently, the report does not claim to provide a complete district-by-district table of exact indicator values or rankings.
- Indicator metadata. Field names were matched to the project variable notes and checked against map patterns. The original metadata was not retained for every field. In particular, the precise denominator for the cultivator share and the original denominator metadata for the supplied WPR fields could not be independently verified.
- Spatial weights. The GeoDa project notes identify Queen contiguity, first-order neighbours and a symmetric weights specification. Sensitivity to alternative weights, such as rook contiguity or distance-based neighbours, was not tested. Spatial results should therefore be interpreted conditional on the selected weights.
- **LISA.** The 5 per cent threshold is not adjusted for multiple testing, and with 35 districts a few significant districts per map could be chance. District identities in the clusters are my reading of unlabelled GeoDa maps, matched to the labelled QGIS maps and checked against the legend counts. Where only one of the two small Mumbai polygons is coloured, I cannot say which.
- **Mumbai.** The Mumbai polygons are too small to check at state scale. They are extreme values for urbanisation and could influence the correlations and Moran's I. I did not test their influence.
- Regression. The models use 35 observations and three explanatory variables. The OLS condition number is high, and the Breusch–Pagan, Koenker–Bassett and White tests indicate heteroskedasticity in the OLS residuals. The OLS residual spatial diagnostics were not available, so the spatial lag/error comparison is not a complete diagnostic-selection exercise. The spatial error model has the lowest AIC among the three models

reported, but this does not establish that it is the only defensible specification. The spatial error pseudo-R^2 is not directly comparable with the ordinary OLS R^2.

- Coverage. Child sex ratio, cultivator share, SC share and ST share were not all subjected to the same full set of spatial statistics. The report distinguishes variables that were mapped from those that were included in Global Moran's I, LISA or regression.
- Map classification. The maps use five classes with unequal intervals, but the classification method was not recorded in the project notes. Colour ramps differ between maps, so the same colour should not be interpreted as the same numerical range across indicators.
- **Ecological and cross-sectional limits.** District averages hide variation inside districts and cannot be applied to individuals. One census year cannot show change, and none of the associations here are causal.

15. References

Census of India. (2011). District Census Handbooks: Maharashtra, Series 28, Part XII-A (Village and Town Directory) and Part XII-B (Primary Census Abstract). Directorate of Census Operations, Maharashtra. Official Census of India catalogue: https://censusindia.gov.in/nada/index.php/catalog/

Anselin, L. (1995). Local Indicators of Spatial Association—LISA. Geographical Analysis, 27(2), 93–115. https://doi.org/10.1111/j.1538-4632.1995.tb00338.x

Anselin, L., Syabri, I., & Kho, Y. (2006). GeoDa: An introduction to spatial data analysis. Geographical Analysis, 38(1), 5–22. https://doi.org/10.1111/j.0016-7363.2005.00671.x

Moran, P. A. P. (1950). Notes on continuous stochastic phenomena. Biometrika, 37(1/2), 17–23. https://doi.org/10.1093/biomet/37.1-2.17

Appendix: Variable Dictionary and Supplementary Analytical Notes

A. Variable dictionary

Field	Indicator	Formula	Unit	Status
MH Indic_4 (Urb_pct)	Urbanisation rate	Urban population / total population × 100	%	Operational definition checked against map; source metadata not retained

Field	Indicator	Formula	Unit	Status
MH Indic_5 (HH_size)	Average household size	Household population / households	Persons	Operational definition checked against map; source metadata not retained
MH Indic_6	Sex ratio	Females / males × 1,000	Females per 1,000 males	Definition interpreted from field and legend; source metadata not retained
MH Indic_7	Child sex ratio	Girls 0–6 / boys 0–6 × 1,000	Girls per 1,000 boys	Not mapped in the main map set
MH Indic_8	Total literacy rate	Literates (7+) / population (7+) × 100	%	Control in regression only
MH Indic11 (Tot_WPR)	Total work participation rate	Workers / population × 100	%	Source denominator not independently verified
MH Indic12	Male work participation rate	Male workers / male population × 100	%	Source metadata not retained; interpreted from field label and map
MH Indic13	Female work participation rate	Female workers / female population × 100	%	Source metadata not retained; interpreted from field label and map
MH Indic14	Cultivators	Share of workers; denominator not independently verified	%	Share rather than count; denominator not independently verified
MH Indic17 (SC_PCT)	Scheduled Caste share	SC population / total population × 100	%	Source metadata not retained; interpreted from field label and

Field	Indicator	Formula	Unit	Status
				map
MH Indic18	Scheduled Tribe share	ST population / total population × 100	%	Source metadata not retained; interpreted from field label and map

B. Supplementary analytical notes

- The district attribute table is not reproduced because a complete cleaned export with all 35 district values was not retained in the report file. The map-based classifications and the GeoDa summary statistics are presented in the relevant sections.
- The working polygon layer is named Maharashtra_District. The original download URL for the boundary file and the exact original Census table identifier were not retained in the project notes. The Census of India District Census Handbooks are cited as the authoritative reference for Census 2011 demographic data.
- Spatial weights used in GeoDa: Queen contiguity, first-order neighbours, symmetric specification; 35 district observations. Alternative weights and sensitivity tests were not estimated.
- The OLS, spatial lag and spatial error estimates are reported in Section 11 and Tables 6 and 6A. OLS Lagrange multiplier diagnostics and Moran's I of OLS residuals were not available in the saved output and are identified as a limitation.
- The cultivator field is interpreted as a share because its map legend is expressed as a percentage, but the denominator was not independently verified. The supplied work-participation fields are analysed as the corresponding WPR indicators, with the source-field denominator limitation stated in Section 14.
- Child sex ratio, cultivator share, SC share and ST share are not presented as having the same complete spatial-statistical treatment as the main indicators. No results are claimed where corresponding outputs were not available.
- The classification method and the geometry of the small Mumbai polygons could not be fully verified from the saved map screenshots. These are stated as limitations rather than treated as confirmed properties.